# A Sheaf-Theoretic Model of Methods

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This paper develops a sheaf-theoretic model of methods. It begins from a primitive account in which methods are purposive progressions of steps in domains of application, producing configurations that may be continued, restricted, compared, or assembled. On that basis it defines presheaves of methods over domains, typed morphisms between methods, strict and weaker forms of descent, and structural transpositions by which methods are relocated between descriptive or operational regimes while remaining comparable under restriction.

In the strict case an abelianization of the coefficients yields a Čech complex of abelian groups, whose augmented quotient furnishes a necessary condition for descent to a single global class. An evaluative branch treats methods whose restriction itself requires adjustment-data, developing the comparison between direct and staged adjusted restriction through weak composites and right weak identity.

Examples illustrate the model in the strict, lax, and fuzzy cases: one for strict compatibility, carrying a presheaf that satisfies the sheaf condition together with two propositions on the failure of assembly, one for successful lax descent, one for failure of lax descent by cocycle breakdown, and a series of fuzzy examples showing successful fuzzy descent, failure of stage-preservation, failure of the triple-overlap law, and a case with common designated structures on the larger domain.

The paper gives a formal development of themes introduced more broadly in *The Science of Methods* by Eric Cafritz (2026), DOI 10.5281/zenodo.18733134, referred to sometimes below as the "Register".

## A Sheaf-Theoretic Model of Methods

## 1. Concept and Rationale

The sheaf-theoretic formalism developed here concerns methods and systems of methods that can be applied in more than one setting, whose different applications can be compared as instances of a continuing procedure, and whose operations may be compared under variation of input and under restriction and extension of domain.

These characteristics permit algebraic, geometric, and sheaf-theoretic modelling of such methods and systems.

This paper develops a formal account of two questions: how local applications of a method assemble over restricted domains, and how methods may be relocated into new descriptive or operational regimes while remaining comparable through restriction.

This paper concerns the modelling of established methods, so that their purposes, steps, and applications are already available for formal comparison, restriction, assembly, and transposition.

The assembly question is illustrated in Section 5 by a coloring of territories, as a full worked example. That example presents a general arrangement, consisting of a finite collection of places at which one act is taken, a set of permitted acts, and a class of complete assignments of acts to places. A coloring map supplies the places as territories and the permitted acts as colors. A timetable assigns the places as the periods of a schedule and the permitted acts as the sessions that may be booked into a period. A complete assignment designates a session for every period.

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# 2. Primitive Objects and Formal Setup

The material in this subsection is foundational. It introduces the primitive objects and distinctions from which the later sheaf-theoretic model is constructed.

Let $U$ be a domain, that is, an expanse in which elements may exist.

A difference in $U$ is any distinction of one of the following kinds: between the presence and absence of an element, between elements, between collections of elements, or between an element and the rest of the domain.

A relation in $U$ is a retained difference.

An object in $U$ is a relation taken as an element of inquiry.

Let $\mathcal{A}(U)$ denote the class of arrangements of relations in $U$.

An arrangement in $U$ is an ordered collection of relations in $U$.

A configuration in $U$ is a present arrangement of relations in $U$, set aside for further study.

A structure in $U$ is a configuration designated for continuation, that is, incorporated into the method as a point of departure for further steps. A configuration may be set aside for further study without serving as a point of departure; it is a structure only when further steps proceed from it. Continuation is further progression from such a configuration.

A step is a deliberate act by which a practitioner alters a configuration in response to one or more selected relations.

Let $\Sigma_U$ denote the set of available steps after the first retained difference in $U$. Let $\Sigma_U^*$ be the free monoid on $\Sigma_U$, whose elements are finite step-sequences, including the empty step-sequence $\varepsilon_U$.

Let $\mathcal{C}(U)$ denote the class of configurations in $U$. A configuration map is a map

$$E_U : \Sigma_U^* \to \mathcal{C}(U)$$

assigning to each step-sequence its resulting configuration. The empty step-sequence $\varepsilon_U$ is assigned the configuration of the empty arrangement in $U$.

Define a relation $\equiv_U$ on $\Sigma_U^*$ by

$$s \equiv_U s' \iff E_U(s) = E_U(s').$$

The quotient

$$Q(U) = \Sigma_U^* / \equiv_U$$

is the configuration quotient of $U$.

Its elements are the $\equiv_U$-equivalence classes of step-sequences, that is, the classes of step-sequences that determine the same configuration under $E_U$, written as subsets $[s]_{\equiv_U} \subseteq \Sigma_U^*$ determined by the relation $\equiv_U$, where

$$[s]_{\equiv_U} = \{\, s' \in \Sigma_U^* : s' \equiv_U s \,\}.$$

Define

$$\mathrm{Quot}_U : Q(U) \to \mathcal{C}(U)$$

by

$$\mathrm{Quot}_U([s]_{\equiv_U}) = E_U(s).$$

This is well-defined by the definition of $\equiv_U$.

The quotient $Q(U)$ records configuration-equivalence at the level of step-sequences, while quotient-configurations are the corresponding configuration-level objects in $\mathcal{C}(U)$. In this paper they are treated as configuration-level objects, but distinguished by their quotient origin through $\mathrm{Quot}_U$.

**Quotient Configurations**

Let $U$ be a domain. A quotient-configuration in $U$ is a configuration in $\mathcal{C}(U)$ determined by an equivalence class $[s]_{\equiv_U} \in Q(U)$.

Equivalently, a quotient-configuration in $U$ is a configuration $c \in \mathcal{C}(U)$ such that

$$E_U(t) = c \qquad \text{for all } t \in [s]_{\equiv_U}$$

for some class $[s]_{\equiv_U} \in Q(U)$.

A quotient-configuration is a configuration viewed through the class of step-sequences that determine it.

A configuration in $\mathcal{C}(U)$ may also be a quotient-configuration: a configuration $c \in \mathcal{C}(U)$ is a quotient-configuration if and only if there exists $s \in \Sigma_U^*$ such that $E_U(s) = c$. A quotient-configuration, once formed, may function as a configuration in subsequent stepwise progression.

The passage from step-sequences to quotient-configurations supports sheaf-theoretic comparison in this model. It identifies local applications by what they determine, rather than by the exact sequence through which that determination is reached.

**Structure Maps**

Let $\mathcal{S}(U) \subseteq \mathcal{C}(U)$ denote the class of structures in $U$, that is, the configurations in $U$ designated for continuation.

Designation is the act by which the method fixes, at a resulting configuration, the structure from which continuation proceeds. The designated structure may be the resulting configuration itself, or a configuration determined by it and retained as the point of departure for further steps.

If $S \subseteq \Sigma_U^*$ is a set of selected step-sequences in $U$, a designation for $S$ consists of a class $D_S \subseteq \{ E_U(s) : s \in S \}$ of configurations at which continuation is designated, together with a designation map

$$\delta_S : D_S \to \mathcal{S}(U)$$

assigning to each such configuration the structure designated for continuation at it. Distinct configurations in $D_S$ may designate the same structure; $\delta_S$ is not required to be injective.

Define

$$Q_S = \{\mathrm{Quot}_U\big([s]_{\equiv_U}\big) : s \in S \text{ and } E_U(s) \in D_S\} \subseteq \mathcal{C}(U).$$

A structure map on $Q_S$ is the assignment

$$\Theta_S : Q_S \to \mathcal{S}(U)$$

given by

$$\Theta_S\big(\mathrm{Quot}_U\big([s]_{\equiv_U}\big)\big) = \delta_S\big(E_U(s)\big).$$

For a selected set $S$ with designation $(D_S, \delta_S)$, the structure map $\Theta_S$ specifies the structure designated for continuation at each quotient-configuration determined by $S$. Where $D_S = \mathcal{S}(U)$ and $\delta_S$ is the inclusion, this reduces to assigning each quotient-configuration its own resulting configuration.

**Proposition — Well-Definedness of the Structure Map**

Let $S \subseteq \Sigma_U^*$ be a set of selected step-sequences in $U$ with designation $(D_S, \delta_S)$. The assignment

$$\Theta_S\big(\mathrm{Quot}_U([s]_{\equiv_U})\big) = \delta_S\big(E_U(s)\big)$$

is well-defined on $Q_S$.

*Proof.* Suppose

$$\mathrm{Quot}_U([s]_{\equiv_U}) = \mathrm{Quot}_U([t]_{\equiv_U}).$$

The quotient-configuration determined by a class is the common resulting configuration of its members, so

$$E_U(s) = E_U(t).$$

Hence $s \equiv_U t$ and $[s]_{\equiv_U} = [t]_{\equiv_U}$, and

$$\delta_S\big(E_U(s)\big) = \delta_S\big(E_U(t)\big).$$

Therefore the value assigned by $\Theta_S$ does not depend on the choice of representative of the quotient-class. Since $Q_S$ contains only quotient-configurations whose resulting configurations lie in $D_S$, the assigned values are defined and lie in $\mathcal{S}(U)$. □

**An Algebra of Relations**

For a domain $U$, let $\mathcal{R}(U)$ denote the class of relations in $U$. Let $\mathcal{R}(U)^*$ denote the class of finite sequences on $\mathcal{R}(U)$, including the empty sequence.

Define the many-sorted algebra

$$\mathbf{Rel}(U) = \big(\Sigma_U^*, \mathcal{R}(U), \mathcal{A}(U), \mathcal{C}(U), Q(U); \mathrm{Concat}_U, \mathrm{Arr}_U, \Gamma_U, E_U, \mathrm{Quot}_U\big),$$

where

$$\mathrm{Concat}_U : \Sigma_U^* \times \Sigma_U^* \to \Sigma_U^*, \qquad \mathrm{Arr}_U : \mathcal{R}(U)^* \to \mathcal{A}(U),$$
$$\Gamma_U : \mathcal{A}(U) \to \mathcal{C}(U), \qquad E_U : \Sigma_U^* \to \mathcal{C}(U), \qquad \mathrm{Quot}_U : Q(U) \to \mathcal{C}(U).$$

Here $\mathrm{Concat}_U$ is concatenation of step-sequences, $\mathrm{Arr}_U$ sends a finite sequence of relations in $U$ to its arrangement in $U$, $\Gamma_U$ sends an arrangement in $U$ to the configuration it determines, $E_U$ sends a step-sequence to its resulting configuration, and $\mathrm{Quot}_U$ sends a class $[s]_{\equiv_U}$ to the quotient-configuration it determines.

This algebra provides the algebraic setting for the sheaf-theoretic formalization that follows.

### Methods and Systems of Methods

A method on $U$ consists of the quintuple

$$m_U = (P_{m_U}, S_{m_U}, C_{m_U}, Q_{m_U}, \Theta_{m_U})$$

together with a designation $(D_{m_U}, \delta_{m_U})$ for its selected step-sequences, where

$$P_{m_U} : \Sigma_U^* \to \{0,1\},$$

$$S_{m_U} = \{\, s \in \Sigma_U^* : P_{m_U}(s) = 1 \,\},$$

$$C_{m_U} = \{\, E_U(s) : s \in S_{m_U} \,\} \subseteq \mathcal{C}(U),$$

and

$$Q_{m_U} = \{\, \mathrm{Quot}_U([s]_{\equiv_U}) : s \in S_{m_U} \text{ and } E_U(s) \in D_{m_U} \,\} \subseteq \mathcal{C}(U),$$

$$\Theta_{m_U} = \Theta_{S_{m_U}} : Q_{m_U} \to \mathcal{S}(U).$$

Here $P_{m_U}$ is the selector for the method on $U$.

The theory assumes that purpose directs the steps of a method; $P_{m_U}$ gives formal expression to that assumption by selecting which step-sequences in $\Sigma_U^*$ belong to the method. The method then includes the selected step-sequences $S_{m_U}$, the configurations $C_{m_U}$ produced by those step-sequences, the quotient-configurations $Q_{m_U}$ determined by those step-sequences, and the structure map $\Theta_{m_U}$, determined by the designation $(D_{m_U}, \delta_{m_U})$. For each quotient-configuration of the method, $\Theta_{m_U}$ assigns the structure designated for continuation at that stage of the method.

A system of methods is a collection of methods, each on its own domain. Systems of methods appear below as raw and covering families of local methods over a common domain.

### Fields of Application and Method-Restriction

An existing method on a domain $U$ applies within at least one field of application in $U$. A field of application of such a method is that part of the domain, fixed by prior designation and constitution, within which selected step-sequences of the method are applied. A method may apply in more than one such field, but for the purposes of this paper one field of application is fixed for each method under consideration.

A subfield of application of a method is a field of application included in the fixed field of application of that method, on which the same method is considered under restriction.

Let

$$m_U = (P_{m_U}, S_{m_U}, C_{m_U}, Q_{m_U}, \Theta_{m_U})$$

be a method on $U$, and let $V \hookrightarrow U$ be a domain whose field of application is a subfield of the fixed field of application of $m_U$. A method

$$m_V = (P_{m_V}, S_{m_V}, C_{m_V}, Q_{m_V}, \Theta_{m_V})$$

on $V$ is a method-restriction of $m_U$ when the selected step-sequences of $m_V$ are selected step-sequences of $m_U$ applied within that subfield.

This is the notion of method-restriction used below.

#### Restriction of Methods, Quotient-Configurations, and Structures

Let $V \hookrightarrow U$ be an inclusion of domains, and let

$$m_U = (P_{m_U}, S_{m_U}, C_{m_U}, Q_{m_U}, \Theta_{m_U})$$

be a method on $U$. Its restriction to $V$ is written

$$\rho_{UV}(m_U) = m_V.$$

Assume there is a restriction map on selected step-sequences

$$\rho^{\Sigma}_{UV} : S_{m_U} \to S_{m_V}.$$

Let

$$\rho^{\mathcal{C}}_{UV} : \mathcal{C}(U) \to \mathcal{C}(V)$$

denote the restriction map on configurations. Restriction of selected step-sequences is required to satisfy

$$E_V(\rho^{\Sigma}_{UV}(s)) = \rho^{\mathcal{C}}_{UV}(E_U(s))$$

for all $s \in S_{m_U}$, and

$$s \equiv_U t \quad \Longrightarrow \quad \rho^{\Sigma}_{UV}(s) \equiv_V \rho^{\Sigma}_{UV}(t)$$

for all $s, t \in S_{m_U}$.

The designation of the restricted method is required to include the configurations induced by restriction:

$$\{\, E_V(\rho^{\Sigma}_{UV}(s)) : s \in S_{m_U} \text{ and } E_U(s) \in D_{m_U} \,\} \subseteq D_{m_V}.$$

In particular, for every $s \in S_{m_U}$ with $E_U(s) \in D_{m_U}$,

$$E_V(\rho^{\Sigma}_{UV}(s)) \in D_{m_V}$$

by this requirement.

The corresponding assignment on quotient classes is

$$[s]_{\equiv_U} \longmapsto [\rho^{\Sigma}_{UV}(s)]_{\equiv_V}.$$

This induces a quotient-configuration restriction map

$$\rho^{Q}_{UV,m_U} : Q_{m_U} \to Q_{m_V}$$

given by

$$\rho^{Q}_{UV,m_U}\big(\mathrm{Quot}_U([s]_{\equiv_U})\big) = \mathrm{Quot}_V([\rho^{\Sigma}_{UV}(s)]_{\equiv_V}).$$

The map is single-valued. Suppose $\mathrm{Quot}_U([s]_{\equiv_U}) = \mathrm{Quot}_U([t]_{\equiv_U})$ for $s, t \in S_{m_U}$. Then $E_U(s) = E_U(t)$ by the definition of $\mathrm{Quot}_U$. Hence $s \equiv_U t$ and $[s]_{\equiv_U} = [t]_{\equiv_U}$. The Well-Definedness Proposition below carries this to $[\rho^{\Sigma}_{UV}(s)]_{\equiv_V} = [\rho^{\Sigma}_{UV}(t)]_{\equiv_V}$, so the two assigned values agree. It lands in $Q_{m_V}$: each element of $Q_{m_U}$ is $\mathrm{Quot}_U([s]_{\equiv_U})$ with $s \in S_{m_U}$ and $E_U(s) \in D_{m_U}$, so $\rho^{\Sigma}_{UV}(s) \in S_{m_V}$ and $E_V(\rho^{\Sigma}_{UV}(s)) \in D_{m_V}$, whence $\mathrm{Quot}_V([\rho^{\Sigma}_{UV}(s)]_{\equiv_V}) \in Q_{m_V}$.

Let

$$\Theta_{m_U} : Q_{m_U} \to \mathcal{S}(U) \qquad \text{and} \qquad \Theta_{m_V} : Q_{m_V} \to \mathcal{S}(V)$$

be the structure maps of $m_U$ and $m_V$, respectively. Restriction is required to satisfy

$$\Theta_{m_V} \circ \rho^{Q}_{UV,m_U} = \rho^{\mathcal{C}}_{UV} \circ \Theta_{m_U}.$$

Both sides are maps $Q_{m_U} \to \mathcal{C}(V)$, and the left side takes values in $\mathcal{S}(V)$. Hence, for every $q \in Q_{m_U}$,

$$\rho^{\mathcal{C}}_{UV}(\Theta_{m_U}(q)) \in \mathcal{S}(V) :$$

restriction of configurations carries the structures designated by the method to structures on the restricted domain. Write $\rho^{\mathcal{S}}_{UV}$ for the restriction of $\rho^{\mathcal{C}}_{UV}$ to those designated structures. No condition is assumed for configurations in $\mathcal{S}(U)$ outside $\Theta_{m_U}(Q_{m_U})$.

**Proposition — Well-Definedness of Restricted Quotient-Class Assignment**

Let $V \hookrightarrow U$ be an inclusion of domains, let

$$m_U = (P_{m_U}, S_{m_U}, C_{m_U}, Q_{m_U}, \Theta_{m_U})$$

be a method on $U$, and let

$$\rho^{\Sigma}_{UV} : S_{m_U} \to S_{m_V}$$

satisfy

$$s \equiv_U t \quad \Longrightarrow \quad \rho^{\Sigma}_{UV}(s) \equiv_V \rho^{\Sigma}_{UV}(t)$$

for all $s, t \in S_{m_U}$.

Then the assignment

$$[s]_{\equiv_U} \longmapsto [\rho^{\Sigma}_{UV}(s)]_{\equiv_V}$$

is well-defined on the subset of quotient classes represented by elements $s \in S_{m_U}$.

*Proof.* Suppose

$$[s]_{\equiv_U} = [t]_{\equiv_U}.$$

Then

$$s \equiv_U t.$$

By hypothesis,

$$\rho^{\Sigma}_{UV}(s) \equiv_V \rho^{\Sigma}_{UV}(t).$$

Therefore

$$[\rho^{\Sigma}_{UV}(s)]_{\equiv_V} = [\rho^{\Sigma}_{UV}(t)]_{\equiv_V}.$$

Thus the assignment depends only on the quotient class represented in $U$ and is well-defined. □

In the tessellation-style examples of Section 5, the equivalence-preservation requirement is met: equality of quotient-configurations is determined by equality of the resulting local tessellation configuration, and restriction to a subdomain preserves that equality.

**Locality Data**

**Restricted Domains and Overlaps**

An inclusion

$$V \hookrightarrow U$$

is the designation of the domain $V$ as included within the domain $U$. In the simplest case, this is inclusion of one domain as a part of another. In other cases, the included domain is a restricted class of cases within the broader domain. For example, the domain of division by single-digit divisors may be included within the broader domain of integer division.

A restricted domain of a domain $U$ is a domain $V$ together with an inclusion

$$V \hookrightarrow U.$$

The inclusion

$$V \hookrightarrow U$$

can only be fully determined by the specifics of the relevant domain on which the method under examination applies.

Let

$$(U_i, m_{U_i}) \to (U, m_U) \qquad \text{and} \qquad (U_j, m_{U_j}) \to (U, m_U)$$

be local objects over a common domain. When the intersection

$$U_i \cap U_j$$

is nonempty and is itself a domain on which the corresponding restricted methods are defined, set

$$U_{ij} := U_i \cap U_j$$

and define $U_{ij}$ as the overlap of $U_i$ and $U_j$.

More generally, for indices $i_0, \dots, i_k$, when the intersection

$$U_{i_0} \cap \cdots \cap U_{i_k}$$

is nonempty and is itself a domain on which the corresponding restricted methods are defined, set

$$U_{i_0\cdots i_k} := U_{i_0} \cap \cdots \cap U_{i_k}$$

and define $U_{i_0\cdots i_k}$ as the $(k+1)$-fold overlap.

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**Strict Branch**

**Sites**

The covering families used in a site of methods are specified according to the form of locality proper to the methods under consideration. The strict form is stated in terms of domains, method-restrictions, and exact agreement on overlaps. Further conditions are developed, when needed, for later descent cases and particular examples.

Let $\mathsf{MethDom}$ denote the collection whose objects are pairs

$$(U, m_U)$$

where $U$ is a domain and $m_U$ is a method on $U$, and whose arrows

$$(V, m_V) \to (U, m_U)$$

consist of an inclusion

$$V \hookrightarrow U$$

and a method-restriction of $m_U$ on $V$, namely $m_V$.

For the strict site, only the primitive restriction data already fixed above are used. In particular, for each inclusion

$$V \hookrightarrow U$$

and each method

$$m_U = (P_{m_U}, S_{m_U}, C_{m_U}, Q_{m_U}, \Theta_{m_U})$$

on $U$, the restriction

$$\rho_{UV}(m_U) = m_V$$

is a method on $V$ together with the corresponding restriction maps on selected step-sequences, quotient-configurations, and structures as introduced above. The strict site will require only those functoriality and domain-closure properties of these restrictions that enter the verification, given below, of the identity axiom, the base-change axiom in the refinement form that passes to the nonempty intersections, and the transitivity axiom.

**Raw Families, Overlap-Closure, and Strict Compatibility**

Let $(U, m_U)$ be an object of $\mathsf{MethDom}$.

A raw family over $(U, m_U)$ is an indexed family of morphisms

$$\Phi = \{(U_i, m_{U_i}) \to (U, m_U)\}_{i \in I}.$$

A raw family

$$\Phi = \{(U_i, m_{U_i}) \to (U, m_U)\}_{i \in I}$$

is overlap-admissible when, for every pair $i, j \in I$ with nonempty intersection

$$U_{ij} := U_i \cap U_j \neq \varnothing,$$

the overlap $U_{ij}$ is itself a domain and the restricted methods

$$\rho_{U_i U_{ij}}(m_{U_i}), \qquad \rho_{U_j U_{ij}}(m_{U_j})$$

are defined.

An overlap-admissible raw family is overlap-closed when, for every pair $i, j \in I$ with nonempty intersection

$$U_{ij} := U_i \cap U_j \neq \varnothing,$$

there is an index $k \in I$ such that

$$U_k = U_{ij}$$

and

$$m_{U_k} = \rho_{U_i U_{ij}}(m_{U_i}) = \rho_{U_j U_{ij}}(m_{U_j}).$$

An overlap-admissible raw family

$$\Phi = \{(U_i, m_{U_i}) \to (U, m_U)\}_{i \in I}$$

is strictly compatible when, for every pair $i, j \in I$ with nonempty intersection

$$U_{ij} := U_i \cap U_j \neq \varnothing,$$

one has exact agreement of the restricted local methods on the overlap:

$$\rho_{U_i U_{ij}}(m_{U_i}) = \rho_{U_j U_{ij}}(m_{U_j}).$$

The strict site uses strict compatibility as its locality condition. Overlap-closure is recorded separately as an auxiliary approach, but is not made part of the covering condition itself.

### Functoriality of Restriction and Domain Closure

Here "functoriality" means only the identity and composition laws for restriction under iterated inclusion, together with the analogous laws for the associated restriction maps on selected step-sequences, quotient-configurations, and structures.

For the strict site, the restriction data are required to satisfy the functoriality laws under iterated inclusion. Whenever

$$W \hookrightarrow V \hookrightarrow U$$

is a chain of inclusions, and

$$m_V = \rho_{UV}(m_U), \qquad m_W = \rho_{VW}(m_V) = \rho_{UW}(m_U),$$

the restriction of methods is required to be path-independent:

$$\rho_{VW}(\rho_{UV}(m_U)) = \rho_{UW}(m_U).$$

Likewise, the corresponding restriction maps on selected step-sequences, quotient-configurations, and structures are required to satisfy the same composition law under iterated inclusion. In particular, for selected step-sequences one requires

$$\rho^{\Sigma}_{UW} = \rho^{\Sigma}_{VW} \circ \rho^{\Sigma}_{UV},$$

for quotient-configurations one requires

$$\rho^{Q}_{UW,m_U} = \rho^{Q}_{VW,m_V} \circ \rho^{Q}_{UV,m_U},$$

and for structures the corresponding law holds on the designated structures without further assumption: for every $q \in Q_{m_U}$, both routes equal $\Theta_{m_W}(\rho^{Q}_{UW,m_U}(q))$ by the structure-map compatibility and the composition law for quotient-configuration restriction,

$$\rho^{\mathcal{S}}_{UW} = \rho^{\mathcal{S}}_{VW} \circ \rho^{\mathcal{S}}_{UV}.$$

The path-independence required above of method-restriction is a condition on the methods being modeled. It excludes those whose restriction to a subdomain depends on the route by which the subdomain is reached. Section 3 takes up that case. There, restriction to a subdomain proceeds through adjustment-data, and the comparison between direct and staged adjusted restriction yields a set of direct adjustment-data rather than a single one.

Finally, the identity restriction is required to act trivially:

$$\rho_{UU}(m_U) = m_U, \qquad \rho^{\Sigma}_{UU} = \mathrm{id}_{S_{m_U}},$$

together with the corresponding identity laws for quotient-configurations and structures.

In addition, the strict site requires closure of domains under nonempty intersection, in the following form. Whenever

$$(V, m_V) \to (U, m_U) \qquad \text{and} \qquad (V', m_{V'}) \to (U, m_U)$$

are morphisms with the same target and the intersection

$$V \cap V'$$

is nonempty, that intersection is required to be a domain, with the corresponding restriction of $m_U$ defined on it.

The intersections used in the verifications below follow from this condition. Where $V \cap V'$ is a domain on which $\rho_{U,V\cap V'}(m_U)$ is defined, the inclusion $V \cap V' \hookrightarrow U$ together with that restricted method is itself a morphism into $(U, m_U)$, so the condition applies again to it and to any further morphism into $(U, m_U)$. By induction, every nonempty intersection of finitely many domains of morphisms into $(U, m_U)$ is a domain on which the corresponding restriction of $m_U$ is defined. The pairwise overlaps $U_i \cap U_j$ of a strict covering family, the intersections $U_i \cap V$ formed under base change together with their own pairwise overlaps $U_i \cap U_j \cap V$, and the intersections of domains drawn from different covering families under transitivity are each of that form.

Under these functoriality laws, with identities given by the identity restrictions and composition given by iterated restriction along chains of inclusions, MethDom is a category.

**Strict Sites**

A strict site is defined for methods whose locality is given by exact agreement of restricted methods on overlaps.

Let $(U, m_U)$ be an object of MethDom. A family of morphisms

$$\Phi = \{(U_i, m_{U_i}) \to (U, m_U)\}_{i \in I}$$

is a strict covering family of $(U, m_U)$ when the following conditions hold.

1. Each
$$(U_i, m_{U_i}) \to (U, m_U)$$
is a morphism in MethDom.
2. The family is overlap-admissible.
3. The family is strictly compatible.
4. The domains of the family jointly cover $U$:
$$U = \bigcup_{i \in I} U_i.$$

Let

$$\mathrm{Cov}_{\mathrm{str}}(U, m_U)$$

denote the class of strict covering families of $(U, m_U)$.

The strict locality condition is therefore exact agreement of restricted methods on pairwise overlaps. No composition law for families is assumed in the definition of strict covering, and no recovery condition for selected step-sequences, quotient-configurations, or structures is built into it.

When needed, one may pass from a strict covering family to an overlap-closed refinement by adjoining the overlap domains together with their common restricted methods. That refinement device is auxiliary and is not part of the definition of strict cover.

Under the functoriality conditions on restriction, together with the closure of the relevant nonempty intersections under pullback and iterated overlap, the strict covering families satisfy the identity axiom, the base-change axiom in the refinement form that passes to the nonempty intersections, and the transitivity axiom. Hence they form a coverage on $\mathsf{MethDom}$ and determine a Grothendieck topology on it.

**Proposition — On Strict Sites**

Assume that method-restriction and the associated restriction maps on selected step-sequences, quotient-configurations, and structures satisfy the functoriality laws under iterated inclusion, and assume further that whenever two morphisms of $\mathsf{MethDom}$ have the same target $(U, m_U)$ and the intersection of their domains is nonempty, that intersection is a domain on which the corresponding restriction of $m_U$ is defined. Then the classes

$$\mathrm{Cov}_{\mathrm{str}}(U, m_U)$$

satisfy the identity axiom, the base-change axiom in the refinement form that passes to the nonempty intersections, and the transitivity axiom, so that they form a coverage on $\mathsf{MethDom}$ and determine a Grothendieck topology on it.

*Proof.* For the identity axiom, the singleton family

$$\{\mathrm{id}_{(U,m_U)} : (U, m_U) \to (U, m_U)\}$$

is a family of morphisms. Its only pair is that of its one member with itself, whose overlap is $U \cap U = U$. This overlap is a domain, and $\rho_{UU}(m_U) = m_U$ is defined on it, so the family is overlap-admissible. The one member agrees with itself there, so the family is strictly compatible. Its only domain is $U$, so the family covers $U$.

For base change, let

$$\Phi = \{(U_i, m_{U_i}) \to (U, m_U)\}_{i \in I}$$

be a strict covering family and let

$$(V, m_V) \to (U, m_U)$$

be any morphism.

The domains $U_i$ and $V$ are the domains of two morphisms with the same target $(U, m_U)$. By the closure condition each nonempty intersection $U_i \cap V$ is a domain on which the corresponding restriction of $m_U$ is defined, and by path-independence of restriction $m_{U_i \cap V} = \rho_{U_i, U_i \cap V}(m_{U_i}) = \rho_{V, U_i \cap V}(m_V)$ is a method-restriction of $m_V$, so one obtains a family of morphisms

$$\Phi|_V := \{(U_i \cap V, m_{U_i \cap V}) \to (V, m_V)\}_{i \in I, U_i \cap V \neq \varnothing}.$$

This family stands as a refinement of $\Phi$, indexed by those $i \in I$ for which the intersection $U_i \cap V$ is nonempty and therefore carries a domain. Each of its members factors through the corresponding member of $\Phi$: the inclusion $U_i \cap V \hookrightarrow U_i$ together with $m_{U_i \cap V} = \rho_{U_i, U_i \cap V}(m_{U_i})$ gives an arrow

$$(U_i \cap V, m_{U_i \cap V}) \to (U_i, m_{U_i}),$$

and the composite of that arrow with $(U_i, m_{U_i}) \to (U, m_U)$ consists of the inclusion $U_i \cap V \hookrightarrow U$ together with the restricted method that path-independence of restriction identifies with $m_{U_i \cap V}$. The composite of $(U_i \cap V, m_{U_i \cap V}) \to (V, m_V)$ with $(V, m_V) \to (U, m_U)$ consists of the same inclusion and the same restricted method, so the two composites are the same arrow. This factorization is what the base-change axiom in the refinement form requires of $\Phi|_V$. The domains of $\Phi|_V$ jointly cover $V$ because the original cover already satisfies $U = \bigcup_{i \in I} U_i$, and hence

$$V = V \cap U = \bigcup_{i \in I} (V \cap U_i).$$

It is overlap-admissible because pairwise overlaps are the nonempty domains

$$(U_i \cap V) \cap (U_j \cap V) = U_i \cap U_j \cap V,$$

and it is strictly compatible by path-independence of restriction together with the strict compatibility of $\Phi$ on $U_i \cap U_j$.

For transitivity, let

$$\Phi = \{(U_i, m_{U_i}) \to (U, m_U)\}_{i \in I}$$

be a strict covering family, and for each $i \in I$ let

$$\Psi_i = \{(V_{ij}, m_{V_{ij}}) \to (U_i, m_{U_i})\}_{j \in J_i}$$

be a strict covering family of $(U_i, m_{U_i})$. Then each composite

$$(V_{ij}, m_{V_{ij}}) \to (U, m_U)$$

is a morphism in $\mathsf{MethDom}$, since by path-independence of restriction $m_{V_{ij}} = \rho_{U_i V_{ij}}(m_{U_i}) = \rho_{U V_{ij}}(m_U)$ is a method-restriction of $m_U$, and the domains jointly cover $U$ because

$$U = \bigcup_{i \in I} U_i = \bigcup_{i \in I} \left( \bigcup_{j \in J_i} V_{ij} \right).$$

Each $V_{ij}$ is the domain of a morphism into $(U, m_U)$, as the preceding step establishes. Any two of them are the domains of two morphisms with the same target, so by the closure condition every nonempty intersection of them is a domain on which the corresponding restriction of $m_U$ is defined. To verify strict compatibility for the composite family, let

$$V_{ij} \cap V_{i'j'} \neq \varnothing.$$

If $i = i'$, then both domains lie inside the same $U_i$, and strict compatibility follows from the strict compatibility of $\Psi_i$ on overlaps within $U_i$. If $i \neq i'$, then the overlap lies inside $U_i \cap U_{i'}$, and strict compatibility of $\Phi$ gives exact agreement of the restricted methods on $U_i \cap U_{i'}$. Restricting further to $V_{ij} \cap V_{i'j'}$, path-independence of restriction carries that agreement to the smaller overlap. In either case, the restricted local methods agree exactly on $V_{ij} \cap V_{i'j'}$. Thus the composite family is again strict covering. $\square$

**Presheaf and Descent**

**Presheaf of Methods**

With the strict locality data and the typed method-comparison data now defined, the descent treatment for the strict, lax, and fuzzy branches is formulated by means of a presheaf assignment. In every branch the local sections are taken in that presheaf. In the strict branch the compatibility data are equalities of sections and are carried by the presheaf as well; in the lax and fuzzy branches the compatibility data are method-comparisons standing over the presheaf on a chosen covering family, with their own conditions under further restriction.

Let

$$\Lambda = \{\mathrm{inc}, \mathrm{res}, \mathrm{ext}, \mathrm{eq}, \mathrm{sqeq}, \mathrm{sqmap}\}$$

index the method-morphism kinds used below: sub-method inclusion, method-restriction, extension, strict equivalence, structure-preserving quotient-configuration bijection, and structure-preserving quotient-configuration map, respectively.

The set $\Lambda$ indexes the method-morphism kinds available for comparing methods. A subset $K \subseteq \Lambda$ is chosen separately for each presheaf, and names the kinds that presheaf tracks functorially. A comparison of a kind indexed by $\Lambda$ therefore compares two methods whether or not that kind belongs to the $K$ of a given presheaf. The directional fuzzy overlap maps introduced below are the quotient-configuration maps underlying comparisons of type sqmap between the restricted local methods, while the lax branch compares restricted local methods through comparisons of type sqeq. The strict branch instead requires literal equality of restricted sections on overlaps, rather than a comparison of type eq.

A set-valued presheaf would be too weak for the present purposes. Since the model must also track selected kinds of morphisms between methods, the target of the presheaf is defined to include both methods and typed classes of method-morphisms.

For a chosen subset $K \subseteq \Lambda$, let $\mathsf{Struct}_K$ denote the category whose objects are collections $X$ of methods together with, for each $k \in K$ and each pair of methods $m, n \in X$, a class

$$\mathrm{Hom}_X^k(m, n)$$

of morphisms of type $k$ from $m$ to $n$.

A morphism $F : X \to Y$ in $\mathsf{Struct}_K$ carries each method $m \in X$ to a method $F(m) \in Y$ and, for each $k \in K$ and each pair $m, n \in X$, gives a map

$$F_{m,n}^k : \mathrm{Hom}_X^k(m, n) \to \mathrm{Hom}_Y^k(F(m), F(n)).$$

Identity morphisms and composition in $\mathsf{Struct}_K$ are given by the identity and composition maps on methods and on each typed class of method-morphisms.

A presheaf of methods of type $K$ on the strict category $\mathsf{MethDom}$ is a functor

$$\mathcal{M} : \mathsf{MethDom}^{op} \to \mathsf{Struct}_K,$$

where $\mathsf{MethDom}^{op}$ denotes the opposite category of $\mathsf{MethDom}$.

To each object $(U, m_U)$ it assigns an object $\mathcal{M}(U, m_U)$ of $\mathsf{Struct}_K$, and to each restriction morphism

$$(V, m_V) \to (U, m_U)$$

arising in the locality data above, it assigns a morphism in $\mathsf{Struct}_K$

$$\rho_{UV} : \mathcal{M}(U, m_U) \to \mathcal{M}(V, m_V),$$

satisfying

$$\rho_{UW} = \rho_{VW} \circ \rho_{UV} \qquad \text{for } (W, m_W) \to (V, m_V) \to (U, m_U), \qquad \rho_{UU} = \mathrm{id}_{\mathcal{M}(U, m_U)}.$$

On the underlying collections of methods, each $\rho_{UV}$ is required to act by method-restriction in the sense fixed above. Thus the underlying collection of $\mathcal{M}(U, m_U)$ consists of methods on $U$, and, for every restriction morphism

$$(V, m_V) \to (U, m_U)$$

and every

$$s \in \mathcal{M}(U, m_U),$$

the method $\rho_{UV}(s)$ is a method-restriction of $s$ on $V$, taken together with the corresponding restriction maps on selected step-sequences, quotient-configurations, and structures. Where more than one method-restriction of $s$ on $V$ is available, the presheaf action selects the one denoted by $\rho_{UV}(s)$; where its corresponding restriction maps are not uniquely determined, those maps are selected as part of the same pointwise restriction data. For every chain

$$W \hookrightarrow V \hookrightarrow U,$$

the direct and staged pointwise restriction data are required to agree under the path-independence and composition laws stated above. On identity inclusions they are required to satisfy the corresponding identity laws.

When convenient, we write

$$s \in \mathcal{M}(U, m_U)$$

for a method in the underlying collection of the $\mathsf{Struct}_K$-object $\mathcal{M}(U, m_U)$. References below to local sections as elements of $\mathcal{M}(U, m_U)$ are to be read as references to that underlying collection.

**Morphisms of Methods**

First let $m_U$ and $n_U$ be methods in the same domain $U$.

A sub-method inclusion of type inc

$$n_U \hookrightarrow m_U$$

is given by

$$S_{n_U} \subseteq S_{m_U}.$$

The method $n_U$ is then a sub-method of $m_U$.

A method-restriction relates methods on different domains. For an inclusion of domains

$$V \hookrightarrow U,$$

a method-restriction of type res

$$m_U \twoheadrightarrow m_V$$

is given by the corresponding restricted method on $V$. In the strict branch,

$$m_V = \rho_{UV}(m_U).$$

In the evaluative branch, the corresponding method on $V$ is obtained by adjusted restriction from $m_U$ relative to specified adjustment-data. In either case, the method-restriction is taken together with the corresponding restriction maps on selected step-sequences, quotient-configurations, and structures.

A strict equivalence of type eq

$$m_U \cong n_U$$

is given by

$$S_{m_U} = S_{n_U}.$$

A structure-preserving quotient-configuration bijection of type sqeq

$$m_U \cong_{SQ} n_U$$

is given by a bijection

$$\phi_Q : Q_{m_U} \to Q_{n_U}$$

such that

$$\Theta_{n_U} \circ \phi_Q = \Theta_{m_U}.$$

A structure-preserving quotient-configuration map of type sqmap

$$m_U \to_{SQ} n_U$$

is given by a map

$$\psi_Q : Q_{m_U} \to Q_{n_U}$$

such that

$$\Theta_{n_U} \circ \psi_Q = \Theta_{m_U}.$$

The defining condition here is the condition on a structure-preserving quotient-configuration bijection with bijectivity dropped, so a bijection satisfying that condition satisfies this one. These maps compose: if

$$\Theta_{n_U} \circ \psi_Q = \Theta_{m_U} \qquad \text{and} \qquad \Theta_{p_U} \circ \chi_Q = \Theta_{n_U},$$

then $\Theta_{p_U} \circ \chi_Q \circ \psi_Q = \Theta_{m_U}$, and the identity map on $Q_{m_U}$ satisfies the condition from $m_U$ to itself.

An extension of type ext

$$n_U \rightsquigarrow m_U$$

holds when $m_U$ is a new method every application of which includes an application of $n_U$. An application of a method is defined as the taking of the steps of that method to a point at which the purpose of the method is deemed to have been met. This may be an obvious point in some methods or it may be a practitioner determined outcome.

A partial composition of methods is an operation

$$m_V \star m_W$$

which, when defined, assigns a method on $V \cup W$ to methods on restricted domains $V$ and $W$.

These morphism types are distinct as comparison relations. The lax and fuzzy descent branches use comparisons of type sqeq and sqmap, respectively, whereas the strict branch uses literal equality of restricted sections on overlaps. Section 5 exhibits methods comparable under sqeq but not under eq.

Inclusion expresses that one method is a sub-method of another.

Method-restriction passes from a method to its restriction on a domain determined by inclusion.

Strict equivalence identifies methods by equality of their selected step-sequences.

A structure-preserving quotient-configuration bijection compares methods whose quotient-configurations correspond bijectively while preserving, through the structure maps, the structures they determine.

A structure-preserving quotient-configuration map compares methods whose quotient-configurations are carried one way, from the first method to the second, preserving through the structure maps the structures they determine. It records a partial fit, since the quotient-configurations of the second method reached by the map are those the first method already determines, while the second method may determine others besides.

Extension preserves a method by including an application of it within a larger application containing further configurations, other methods, or both.

## Descent Cases

### Descent

Descent concerns the passage from local sections on a covering family to a global section on the larger domain. It is stated first in the strict case, where the passage is defined by equality of the restricted local sections on overlaps. Weaker forms of descent, in which overlap comparison is given by weaker morphisms, are introduced in the subsequent section.

### Sheaf Condition (Strict)

For a strict covering family

$$\{(U_i, m_{U_i}) \to (U, m_U)\},$$

write

$$U_{ij} := U_i \cap U_j$$

for each nonempty overlap, with $m_{U_{ij}} := \rho_{UU_{ij}}(m_U)$. By path-independence of restriction, $\rho_{U_iU_{ij}}(m_{U_i}) = \rho_{UU_{ij}}(m_U) = \rho_{U_jU_{ij}}(m_{U_j})$, so the restricted local sections below lie in the common object $\mathcal{M}(U_{ij}, m_{U_{ij}})$.

Let

$$s_i \in \mathcal{M}(U_i, m_{U_i})$$

be local sections. The strict sheaf condition is that if

$$\rho_{U_iU_{ij}}(s_i) = \rho_{U_jU_{ij}}(s_j) \quad \text{in } \mathcal{M}(U_{ij}, m_{U_{ij}})$$

for every pair $i, j$ with nonempty overlap, then there exists a unique

$$s \in \mathcal{M}(U, m_U)$$

such that

$$\rho_{UU_i}(s) = s_i$$

for each $i$.

Exact agreement on overlaps is the strict compatibility criterion. Assembly requires in addition that the presheaf satisfy the strict sheaf condition. Section 5 describes a presheaf of methods satisfying the condition, where assembly holds in existence and in uniqueness, together with families for which each part fails. Comparison conditions on overlaps are developed alongside assembly, since those conditions stand where assembly fails.

**Proposition — Strict Assembly Criterion**

Let

$$\mathcal{M} : \mathsf{MethDom}^{op} \to \mathsf{Struct}_K$$

be a presheaf of methods satisfying the strict sheaf condition. Let

$$\{(U_i, m_{U_i}) \to (U, m_U)\}$$

be a strict covering family, and let

$$s_i \in \mathcal{M}(U_i, m_{U_i})$$

be local sections.

Write

$$U_{ij} := U_i \cap U_j$$

for each nonempty overlap.

Then the following are equivalent.

1. There exists a global section

   $$s \in \mathcal{M}(U, m_U)$$

   such that

   $$\rho_{UU_i}(s) = s_i$$

   for each $i$.

2. For every pair $i, j$ with nonempty overlap, the restricted local sections agree on the overlap:

   $$\rho_{U_iU_{ij}}(s_i) = \rho_{U_jU_{ij}}(s_j).$$

When these conditions hold, the global section $s$ is unique.

*Proof.* Assume that there exists a global section

$$s \in \mathcal{M}(U, m_U)$$

such that

$$\rho_{UU_i}(s) = s_i$$

for each $i$. Let
$$U_{ij} := U_i \cap U_j$$
for each nonempty overlap. By functoriality of restriction,

$$\rho_{U_iU_{ij}}(s_i) = \rho_{U_iU_{ij}}(\rho_{UU_i}(s)) = \rho_{UU_{ij}}(s).$$

Similarly,
$$\rho_{U_jU_{ij}}(s_j) = \rho_{U_jU_{ij}}(\rho_{UU_j}(s)) = \rho_{UU_{ij}}(s).$$

Hence
$$\rho_{U_iU_{ij}}(s_i) = \rho_{U_jU_{ij}}(s_j)$$
for every pair $i, j$ with nonempty overlap.

Conversely, assume that for every pair $i, j$ with nonempty overlap,

$$\rho_{U_iU_{ij}}(s_i) = \rho_{U_jU_{ij}}(s_j).$$

Since $\mathcal{M}$ satisfies the strict sheaf condition, there exists a unique section

$$s \in \mathcal{M}(U, m_U)$$

such that
$$\rho_{UU_i}(s) = s_i$$
for each $i$.

This establishes that a global section with the stated restrictions exists exactly when the local sections agree on every nonempty overlap. Uniqueness then follows from the strict sheaf condition.

□

**Lax and Fuzzy Descent**

In the strict, lax, and fuzzy cases, local sections are still taken in the presheaf $\mathcal{M}$. What changes is the compatibility data used on overlaps. In the strict case, compatibility is equality after restriction. In the lax case, compatibility is carried by comparisons of type sqeq between the restricted local methods on each overlap. In the fuzzy case, it is carried by comparisons of type sqmap between them. In both cases the comparisons stand over the presheaf on the chosen covering family, and their behavior under further restriction is stated as a condition on the descent datum rather than supplied by the presheaf.

For each object $(U, m_U)$, the object

$$\mathcal{M}(U, m_U)$$

of $\mathsf{Struct}_K$ has an underlying collection consisting of methods on $U$ and carries, for each chosen morphism kind $k \in K$, the corresponding typed comparison classes

$$\mathrm{Hom}^k_{\mathcal{M}(U,m_U)}(m, n).$$

For each restriction morphism

$$(V, m_V) \to (U, m_U),$$

the restriction map

$$\rho_{UV} : \mathcal{M}(U, m_U) \to \mathcal{M}(V, m_V)$$

has the following action.

On the underlying collection of methods, this map sends each

$$s \in \mathcal{M}(U, m_U)$$

to the method-restriction $\rho_{UV}(s)$ of $s$ on $V$ selected by the presheaf action. The selected pointwise restriction data include the corresponding restriction maps on selected step-sequences, quotient-configurations, and structures. For each $k \in K$ and each pair

$$m, n \in \mathcal{M}(U, m_U),$$

the map also gives a map

$$\mathrm{Hom}^k_{\mathcal{M}(U,m_U)}(m, n) \longrightarrow \mathrm{Hom}^k_{\mathcal{M}(V,m_V)}\big(\rho_{UV}(m), \rho_{UV}(n)\big).$$

The lax and fuzzy branches are formulated through typed overlap-comparison data standing over this presheaf on a chosen covering family. On each overlap, the restricted local methods are compared by quotient-configuration bijections in the lax case and by quotient-configuration maps in the directional fuzzy case. A comparison on a pairwise overlap leaves the corresponding comparison on a triple overlap contained in it undetermined, since a map making the restriction square commute exists only in some cases, and where it exists there may be several. The descent datum therefore carries a chosen comparison on each triple overlap, together with the requirement that the restriction square from each containing pairwise overlap commute, and composites on triple overlaps are taken among the chosen maps. In the strict case, equality of restricted local sections on pairwise overlaps already determines compatibility on triple overlaps. In the lax and fuzzy cases, triple overlaps test whether the weaker comparison data fit together consistently.

**Lax Descent**

In the lax case, overlap comparison is made through structure-preserving quotient-configuration bijections between the quotient-configurations of the restricted local methods.

For a family

$$\{(U_i, m_{U_i}) \to (U, m_U)\},$$

let

$$s_i \in \mathcal{M}(U_i, m_{U_i})$$

be local sections. For each nonempty overlap

$$U_{ij} := U_i \cap U_j,$$

let

$$m_i^{ij} := \rho_{U_i U_{ij}}(s_i), \qquad m_j^{ij} := \rho_{U_j U_{ij}}(s_j)$$

denote the restricted local methods on $U_{ij}$. Write

$$Q_i^{ij} := Q_{m_i^{ij}}, \qquad Q_j^{ij} := Q_{m_j^{ij}}$$

for their quotient-configuration sets, and

$$\Theta_i^{ij} : Q_i^{ij} \to \mathcal{S}(U_{ij}), \qquad \Theta_j^{ij} : Q_j^{ij} \to \mathcal{S}(U_{ij})$$

for their corresponding structure maps. A lax overlap comparison on $U_{ij}$ is the quotient-configuration bijection underlying an sqeq-morphism between the restricted local methods, namely a bijection

$$\alpha_{ij} : Q_i^{ij} \to Q_j^{ij}$$

such that

$$\Theta_j^{ij} \circ \alpha_{ij} = \Theta_i^{ij}.$$

Now let

$$U_{ijk} := U_i \cap U_j \cap U_k$$

be a nonempty triple overlap. Define the further restricted local methods

$$m_i^{ijk} := \rho_{U_i U_{ijk}}(s_i), \qquad m_j^{ijk} := \rho_{U_j U_{ijk}}(s_j), \qquad m_k^{ijk} := \rho_{U_k U_{ijk}}(s_k).$$

By functoriality of restriction, these are the same local methods obtained by restricting the corresponding pairwise overlap methods further to $U_{ijk}$.

Write

$$Q_i^{ijk} := Q_{m_i^{ijk}}, \qquad Q_j^{ijk} := Q_{m_j^{ijk}}, \qquad Q_k^{ijk} := Q_{m_k^{ijk}}$$

for the quotient-configuration sets of these further restricted local methods on $U_{ijk}$, and

$$\Theta_i^{ijk} : Q_i^{ijk} \to \mathcal{S}(U_{ijk}), \qquad \Theta_j^{ijk} : Q_j^{ijk} \to \mathcal{S}(U_{ijk}), \qquad \Theta_k^{ijk} : Q_k^{ijk} \to \mathcal{S}(U_{ijk})$$

for their corresponding structure maps. A lax overlap comparison $\alpha_{ij}$ is said to restrict to the triple overlap when there exists a bijection

$$\alpha_{ij}^{ijk} : Q_i^{ijk} \to Q_j^{ijk}$$

such that

$$\rho^{Q}_{U_{ij} U_{ijk}, m_j^{ij}} \circ \alpha_{ij} = \alpha_{ij}^{ijk} \circ \rho^{Q}_{U_{ij} U_{ijk}, m_i^{ij}},$$

and

$$\Theta_j^{ijk} \circ \alpha_{ij}^{ijk} = \Theta_i^{ijk}.$$

Define $\alpha_{jk}^{ijk}$ and $\alpha_{ik}^{ijk}$ similarly.

Fix a linear order on the index set $I$. Lax overlap comparisons are supplied for the ordered pairs $i < j$ alone, and the condition below is stated for triples $i < j < k$. Each such comparison is a bijection, and its inverse satisfies the same structure-map condition, so the comparison in the other direction is available from it.

A lax descent datum is a family of lax overlap comparisons

$$\{\alpha_{ij}\}_{i<j}$$

together with, for every nonempty triple overlap $U_{ijk}$ with $i < j < k$, a chosen structure-preserving bijection on $U_{ijk}$ for each of the three pairs, each restricting the corresponding pairwise overlap comparison in the preceding sense, such that the chosen bijections satisfy the cocycle condition

$$\alpha_{jk}^{ijk} \circ \alpha_{ij}^{ijk} = \alpha_{ik}^{ijk}.$$

The chosen bijections belong to the datum because a pairwise overlap comparison may restrict to a triple overlap in more than one way. Where more than one bijection satisfies the restriction square together with the structure-map condition, the composite $\alpha_{jk}^{ijk} \circ \alpha_{ij}^{ijk}$ and the direct bijection $\alpha_{ik}^{ijk}$ both depend on which of them the datum carries. The selection determines whether the cocycle condition is satisfied, and a family of pairwise overlap comparisons taken by itself leaves the condition open. Such a family constitutes a lax descent datum when some selection of restricted bijections satisfies the cocycle condition on every nonempty triple overlap $U_{ijk}$ with $i < j < k$.

Where the restriction map

$$\rho^{Q}_{U_{ij}U_{ijk},m_i^{ij}} : Q_i^{ij} \to Q_i^{ijk}$$

is surjective, at most one map satisfies the restriction square for the pair $i, j$, so the pairwise overlap comparison $\alpha_{ij}$ determines the restricted bijection the datum carries on $U_{ijk}$. The proposition establishing this appears below in the fuzzy branch, stated for structure-preserving maps, and a structure-preserving quotient-configuration bijection is such a map.

**The Fuzzy Case** In the fuzzy case, the fit between local methods on an overlap is allowed to be partial or approached rather than exact. The fuzzy branch identifies what is preserved, under such partial fit, at the level of quotient-configurations and structures.

### Fuzzy Comparison

Overlap comparison is first taken here in a symmetric sense. Two restricted local methods are compared on an overlap at the level of quotient-configurations and of the configurations designated for continuation by them, without yet requiring either exact agreement of the restricted local methods or bijective correspondence of quotient-configurations.

For a family

$$\{(U_i, m_{U_i}) \to (U, m_U)\},$$

let

$$s_i \in \mathcal{M}(U_i, m_{U_i})$$

be local sections. For each nonempty overlap

$$U_{ij} := U_i \cap U_j,$$

let

$$m_i^{ij} := \rho_{U_i U_{ij}}(s_i), \qquad m_j^{ij} := \rho_{U_j U_{ij}}(s_j)$$

denote the restricted local methods on $U_{ij}$. Write

$$Q_i^{ij} := Q_{m_i^{ij}}, \qquad Q_j^{ij} := Q_{m_j^{ij}}$$

for their quotient-configuration sets, and

$$\Theta_i^{ij} : Q_i^{ij} \to \mathcal{S}(U_{ij}), \qquad \Theta_j^{ij} : Q_j^{ij} \to \mathcal{S}(U_{ij})$$

for their corresponding structure maps.

A fuzzy comparison on $U_{ij}$ is given when the two restricted local methods share at least one designated structure on the overlap:

$$\Theta_i^{ij}(Q_i^{ij}) \cap \Theta_j^{ij}(Q_j^{ij}) \neq \varnothing.$$

Write

$$m_i^{ij} \overset{\text{fuzz}}{\sim} m_j^{ij}$$

when this condition holds.

This relation expresses partial fit at the level of the configurations designated for continuation: some structure designated by one restricted local method on the overlap is also designated by the other. At this level, no preferred direction of comparison is fixed, and fuzzy descent has not yet been imposed.

### Fuzzy Descent

Fuzzy descent introduces directional transport maps and the triple-overlap law.

The symmetric relation $m_i^{ij} \overset{\text{fuzz}}{\sim} m_j^{ij}$ is called fuzzy comparison. The directional maps introduced next are called fuzzy overlap comparisons.

Retain the notation $m_i^{ij}$, $m_j^{ij}$, $Q_i^{ij}$, $Q_j^{ij}$, $\Theta_i^{ij}$, and $\Theta_j^{ij}$ introduced above.

A fuzzy overlap comparison on $U_{ij}$ is the quotient-configuration map underlying a comparison of type sqmap between the restricted local methods $m_i^{ij}$ and $m_j^{ij}$, that is, a map

$$\beta_{ij}^{Q} : Q_i^{ij} \to Q_j^{ij}$$

such that

$$\Theta_j^{ij} \circ \beta_{ij}^{Q} = \Theta_i^{ij}.$$

Thus the quotient-configurations on one side of the overlap are carried to those on the other side in a way that preserves the configurations designated for continuation on the overlap.

**Proposition — A Directional Comparison Carries the Symmetric One**

Let $m_U$ and $n_U$ be methods in the same domain $U$, and let

$$\psi_Q : Q_{m_U} \to Q_{n_U}$$

be a structure-preserving quotient-configuration map. Then

$$\Theta_{m_U}(Q_{m_U}) \subseteq \Theta_{n_U}(Q_{n_U}).$$

If in addition $Q_{m_U}$ is nonempty, then

$$\Theta_{m_U}(Q_{m_U}) \cap \Theta_{n_U}(Q_{n_U}) \neq \varnothing.$$

*Proof.* Let $q \in Q_{m_U}$. By the defining condition,

$$\Theta_{m_U}(q) = \Theta_{n_U}(\psi_Q(q)),$$

which lies in $\Theta_{n_U}(Q_{n_U})$. Hence $\Theta_{m_U}(Q_{m_U}) \subseteq \Theta_{n_U}(Q_{n_U})$.

Suppose further that $Q_{m_U}$ is nonempty and take $q \in Q_{m_U}$. Then $\Theta_{m_U}(q)$ belongs to $\Theta_{m_U}(Q_{m_U})$ and, by the containment just established, to $\Theta_{n_U}(Q_{n_U})$ as well, so the two images share at least one designated structure. □

Applied on an overlap $U_{ij}$ to the restricted local methods $m_i^{ij}$ and $m_j^{ij}$, the proposition gives

$$m_i^{ij} \overset{\text{fuzz}}{\sim} m_j^{ij}$$

whenever a fuzzy overlap comparison is available on that overlap and $Q_i^{ij}$ is nonempty. The directional case therefore carries the symmetric one under that hypothesis, and the hypothesis is what the passage requires: a comparison out of an empty quotient-configuration set preserves the designated structures with nothing to carry, and leaves the two images sharing nothing.

Now let

$$U_{ijk} := U_i \cap U_j \cap U_k$$

be a nonempty triple overlap. Define the further restricted local methods

$$m_i^{ijk} := \rho_{U_i U_{ijk}}(s_i), \qquad m_j^{ijk} := \rho_{U_j U_{ijk}}(s_j), \qquad m_k^{ijk} := \rho_{U_k U_{ijk}}(s_k).$$

Write

$$Q_i^{ijk} := Q_{m_i^{ijk}}, \qquad Q_j^{ijk} := Q_{m_j^{ijk}}, \qquad Q_k^{ijk} := Q_{m_k^{ijk}}$$

for the quotient-configuration sets of these further restricted local methods on $U_{ijk}$, and

$$\Theta_i^{ijk} : Q_i^{ijk} \to \mathcal{S}(U_{ijk}), \qquad \Theta_j^{ijk} : Q_j^{ijk} \to \mathcal{S}(U_{ijk}), \qquad \Theta_k^{ijk} : Q_k^{ijk} \to \mathcal{S}(U_{ijk})$$

for their corresponding structure maps.

A fuzzy overlap comparison $\beta_{ij}^Q$ is said to restrict to the triple overlap when there exists a map

$$\beta_{ij}^{Q,ijk} : Q_i^{ijk} \to Q_j^{ijk}$$

such that

$$\rho^Q_{U_{ij}U_{ijk}, m_j^{ij}} \circ \beta_{ij}^Q = \beta_{ij}^{Q,ijk} \circ \rho^Q_{U_{ij}U_{ijk}, m_i^{ij}},$$

and

$$\Theta_j^{ijk} \circ \beta_{ij}^{Q,ijk} = \Theta_i^{ijk}.$$

Define $\beta_{jk}^{Q,ijk}$ and $\beta_{ik}^{Q,ijk}$ similarly.

Fix a linear order on the index set $I$. Fuzzy overlap comparisons are supplied for the ordered pairs $i < j$ alone, and the law below is stated for triples $i < j < k$. The orientation belongs to the datum. Were comparisons supplied for every ordered pair together with identity comparisons, the law applied to the triple $i, j, i$, whose triple overlap is $U_{ij}$, would give $\beta_{ji}^Q \circ \beta_{ij}^Q = \mathrm{id}$, and likewise in the other order, so every comparison would be a bijection and the directional case would return to the lax one.

A fuzzy descent datum is a family of fuzzy overlap comparisons

$$\{\beta_{ij}^Q\}_{i<j}$$

together with, for every nonempty triple overlap $U_{ijk}$ with $i < j < k$, a chosen structure-preserving map on $U_{ijk}$ for each of the three pairs, each restricting the corresponding pairwise overlap comparison in the preceding sense, such that the chosen maps satisfy

$$\beta_{jk}^{Q,ijk} \circ \beta_{ij}^{Q,ijk} = \beta_{ik}^{Q,ijk}.$$

The chosen maps belong to the datum because a pairwise overlap comparison may restrict to a triple overlap in more than one way. Where more than one map satisfies the restriction square together with the structure-map condition, the composite $\beta^{Q,ijk}_{jk} \circ \beta^{Q,ijk}_{ij}$ and the direct map $\beta^{Q,ijk}_{ik}$ both depend on which of them the datum carries. The selection determines whether the triple-overlap law is satisfied, and a family of fuzzy overlap comparisons taken by itself leaves the law open. Such a family constitutes a fuzzy descent datum when some selection of restricted maps satisfies the triple-overlap law on every nonempty triple overlap $U_{ijk}$ with $i < j < k$.

**Proposition — A Surjective Source Restriction Determines the Restricted Comparison**

Let $U_{ij}$ be a nonempty pairwise overlap containing a nonempty triple overlap $U_{ijk}$, retain the notation above, and write

$$r_i := \rho^{Q}_{U_{ij}U_{ijk},\, m_i^{ij}} : Q_i^{ij} \to Q_i^{ijk}, \qquad r_j := \rho^{Q}_{U_{ij}U_{ijk},\, m_j^{ij}} : Q_j^{ij} \to Q_j^{ijk}$$

for the restriction maps to the triple overlap. Let $\beta^Q_{ij}$ be a fuzzy overlap comparison on $U_{ij}$, and suppose that $r_i$ is surjective. Then any two maps

$$\gamma, \gamma' : Q_i^{ijk} \to Q_j^{ijk}$$

satisfying the restriction square

$$r_j \circ \beta^Q_{ij} = \gamma \circ r_i, \qquad r_j \circ \beta^Q_{ij} = \gamma' \circ r_i$$

are equal. Under this hypothesis the pairwise overlap comparison $\beta^Q_{ij}$ determines the map $\beta^{Q,ijk}_{ij}$ that a descent datum carries for the pair $i, j$ on $U_{ijk}$.

*Proof.* Let $q \in Q_i^{ijk}$. Since $r_i$ is surjective, there is $z \in Q_i^{ij}$ with $r_i(z) = q$. The two restriction squares give

$$\gamma(q) = \gamma\big(r_i(z)\big) = r_j\big(\beta^Q_{ij}(z)\big) = \gamma'\big(r_i(z)\big) = \gamma'(q).$$

The two maps agree at each element of $Q_i^{ijk}$, so $\gamma = \gamma'$. □

The argument proceeds from the restriction square alone, and holds for any two maps satisfying it. It therefore applies in the lax branch as well, where the comparisons are structure-preserving bijections and each such bijection is a structure-preserving map. Surjectivity of $r_i$ yields uniqueness of the restricted comparison; existence remains a separate question, settled in a given case by exhibiting a map that satisfies the restriction square together with the structure-map condition. Where $r_i$ reaches part of $Q_i^{ijk}$, the square leaves the values on the remainder open, and the selection carried by the datum may bear on the triple-overlap law.

The preceding subsection shows symmetric fuzzy comparison through shared designated structures on each overlap. The present subsection introduces the directional transport maps and the triple-overlap law by which a fuzzy descent datum is constituted.

Examples of fuzzy comparison and fuzzy descent are given in Section 5 below.

**Directional Fuzzy Overlap Comparisons**

The lax overlap comparison defined above is the quotient-configuration bijection underlying an sqeq-morphism. It matches the quotient-configurations of one restricted local method to those of the other, and it is reversible: the inverse bijection satisfies the same structure-map condition. The fuzzy overlap comparison relaxes the bijection to a structure-preserving map $\beta^Q_{ij} : Q^{ij}_i \to Q^{ij}_j$ with $\Theta^{ij}_j \circ \beta^Q_{ij} = \Theta^{ij}_i$. Such a map may identify distinct source configurations in a single target configuration, and may take its values in part of $Q^{ij}_j$. Where distinct sources share an image, a single map returns only one of them from that image, so no left inverse exists. Where a target configuration lies outside the image, no source is sent to it, so no right inverse exists. A structure-preserving two-sided inverse is therefore available exactly where the comparison is bijective, and in that case the comparison is the quotient-configuration bijection underlying an sqeq-morphism of the restricted local methods. Outside that case the comparison carries configurations in one direction only.

**Čech Viewpoints**

**Nonabelian Čech Viewpoint for Lax Descent**

Retain the linear order on $I$ fixed above. For each nonempty overlap $U_{ij} := U_i \cap U_j$ with $i < j$, let

$$\alpha_{ij} : Q^{ij}_i \to Q^{ij}_j$$

be a structure-preserving quotient-configuration bijection, so that

$$\Theta^{ij}_j \circ \alpha_{ij} = \Theta^{ij}_i.$$

When the source and target quotient-configuration sets are the same, the corresponding structure-preserving automorphisms are

$$\mathrm{Aut}_\Theta(Q) := \{\sigma : Q \to Q \mid \sigma \text{ is a bijection and } \Theta \circ \sigma = \Theta\}.$$

In general, however, the maps $\alpha_{ij}$ need not be automorphisms of one fixed quotient-configuration set. They are structure-preserving transition bijections between the quotient-configuration sets of the restricted local methods. A lax descent datum may be viewed as nonabelian Čech 1-cochain data formed by these transition bijections, or as groupoid-valued overlap data with the relevant quotient-configuration sets as objects.

Assume that on each triple overlap $U_{ijk}$, the pairwise overlap bijections restrict to corresponding sqeq-bijections

$$\alpha^{ijk}_{ij} : Q^{ijk}_i \to Q^{ijk}_j, \qquad \alpha^{ijk}_{jk} : Q^{ijk}_j \to Q^{ijk}_k, \qquad \alpha^{ijk}_{ik} : Q^{ijk}_i \to Q^{ijk}_k,$$

where $Q_i^{ijk}, Q_j^{ijk}, Q_k^{ijk}$ are the quotient-configuration sets of the restricted local methods on $U_{ijk}$. Such a family is a lax descent cocycle when

$$\alpha_{jk}^{ijk} \circ \alpha_{ij}^{ijk} = \alpha_{ik}^{ijk}$$

on every nonempty triple overlap $U_{ijk}$ with $i < j < k$.

When this condition fails, the restricted bijections assumed above do not constitute a lax descent datum. Whether the pairwise overlap comparisons themselves fail depends on the restriction maps from the pairwise overlaps to the triple overlap. Where those maps are surjective, the proposition above determines the restricted bijections from the pairwise overlap comparisons. No other selection of restricted bijections is then available, and the pairwise overlap comparisons themselves fail. Where those maps are not surjective, some other selection may satisfy the cocycle condition, and the failure belongs to the restricted bijections assumed above rather than to the pairwise overlap comparisons.

### The Fuzzy Data from the Čech Viewpoint

The nonabelian lax construction presents overlap data through transition comparisons that are invertible; every transition arrow in the coefficient system is reversible.

The fuzzy overlap comparisons proceed through directional maps. On each triple overlap the restricted comparisons compose as structure-preserving maps, and the triple-overlap law $\beta_{jk}^{Q,ijk} \circ \beta_{ij}^{Q,ijk} = \beta_{ik}^{Q,ijk}$ equates the composite of the restricted comparisons with the direct restricted comparison. Where the relevant directional comparisons are bijective, the reversible lax Čech viewpoint applies to the fuzzy data. Outside the bijective case, the directional comparison maps compose without requiring inverses, and a Čech treatment of that case would require a coefficient system defined over composable structure-preserving maps rather than over invertible ones.

### Strict Čech Construction

The nonabelian viewpoints above carry overlap data through transition comparisons. Where the coefficients instead have an abelianization, the strict case supports a Čech complex of abelian groups, and non-trivial cohomology signals an obstruction to a global section. This is the case described in Annex I of the Register for linear coefficients, and the construction below discharges it for strict descent.

We assume that the underlying collections of methods in the presheaf $\mathcal{M}$ are sets, so that the free abelian group construction applies.

Assume that $K$ contains the morphism-kind eq of strict equivalence. For each object $(U, m_U)$, let

$$\overline{\mathcal{M}}(U, m_U)$$

denote the set of strict-equivalence classes of methods in $\mathcal{M}(U, m_U)$, and write

$$[s]_{\mathrm{eq}}$$

for the strict-equivalence class of a method $s \in \mathcal{M}(U, m_U)$. Define

$$\mathcal{G}(U, m_U) := \mathbb{Z}[\overline{\mathcal{M}}(U, m_U)],$$

the free abelian group on $\overline{\mathcal{M}}(U, m_U)$.

Each restriction map

$$\rho_{UV} : \mathcal{M}(U, m_U) \longrightarrow \mathcal{M}(V, m_V)$$

preserves strict equivalence, and so induces a well-defined map

$$\overline{\rho}_{UV} : \overline{\mathcal{M}}(U, m_U) \longrightarrow \overline{\mathcal{M}}(V, m_V)$$

given by

$$\overline{\rho}_{UV}([s]_{\mathrm{eq}}) := [\rho_{UV}(s)]_{\mathrm{eq}}.$$

This in turn induces a homomorphism of abelian groups

$$\rho^{\mathcal{G}}_{UV} := \mathbb{Z}[\overline{\rho}_{UV}] : \mathcal{G}(U, m_U) \longrightarrow \mathcal{G}(V, m_V)$$

by linear extension. The functoriality of restriction on methods yields functoriality of these induced maps, so that $\mathcal{G}$ is a presheaf of abelian groups on $\mathsf{MethDom}$.

Let

$$\mathfrak{U} = \{(U_i, m_{U_i}) \to (U, m_U)\}_{i \in I}$$

be a strict covering family, and retain the linear order on the index set $I$. For indices $i_0 < \cdots < i_k$ such that

$$U_{i_0} \cap \cdots \cap U_{i_k} \neq \varnothing,$$

write

$$U_{i_0 \cdots i_k} := U_{i_0} \cap \cdots \cap U_{i_k},$$

and let

$$m_{U_{i_0 \cdots i_k}} := \rho_{U, U_{i_0 \cdots i_k}}(m_U).$$

Under the domain-closure hypothesis used for the strict site, each such nonempty finite intersection is a domain on which this restricted method is defined. The group of $k$-cochains is

$$C^k(\mathfrak{U}, \mathcal{G}) := \prod_{\substack{i_0 < \cdots < i_k \\ U_{i_0 \cdots i_k} \neq \varnothing}} \mathcal{G}\big(U_{i_0 \cdots i_k}, m_{U_{i_0 \cdots i_k}}\big).$$

For a $k$-cochain $c$ and every $i_0 < \cdots < i_{k+1}$ with $U_{i_0 \cdots i_{k+1}} \neq \varnothing$, define

$$(dc)_{i_0 \cdots i_{k+1}} := \sum_{r=0}^{k+1} (-1)^r \, \rho^{\mathcal{G}}_{U_{i_0 \cdots \widehat{i_r} \cdots i_{k+1}}, U_{i_0 \cdots i_{k+1}}} \big(c_{i_0 \cdots \widehat{i_r} \cdots i_{k+1}}\big).$$

Functoriality of the induced restriction maps gives

$$d^2 = 0.$$

In particular, for a 0-cochain $c = (c_i)$,

$$(dc)_{ij} = \rho^{\mathcal{G}}_{U_j U_{ij}}(c_j) - \rho^{\mathcal{G}}_{U_i U_{ij}}(c_i) \qquad (i < j).$$

A 0-cocycle is a family $(c_i)$ with

$$(dc)_{ij} = 0$$

for all $i < j$ with nonempty overlap. A general 0-cocycle is a family of formal integral linear combinations of the basis generators indexed by strict-equivalence classes that are compatible under restriction. A special case consists of generator-valued 0-cocycles, where

$$c_i = \mathbf{e}_{[s_i]_{\mathrm{eq}}}$$

is the basis generator associated with the strict-equivalence class of some local method

$$s_i \in \mathcal{M}(U_i, m_{U_i}).$$

**Proposition — Strict Local Data Determine a Generator-Valued Čech 0-Cocycle**

Assume that $K$ contains eq, and let $\mathcal{G}$ be the presheaf of abelian groups defined above. Let $\mathfrak{U}$ be a strict covering family with a fixed order on $I$, and let

$$s_i \in \mathcal{M}(U_i, m_{U_i})$$

be local sections. If

$$\rho_{U_i U_{ij}}(s_i) = \rho_{U_j U_{ij}}(s_j)$$

for every pair $i, j$ with nonempty overlap, then setting

$$c_i := \mathbf{e}_{[s_i]_{\mathrm{eq}}} \in \mathcal{G}(U_i, m_{U_i})$$

yields a generator-valued 0-cocycle

$$c = (c_i) \in C^0(\mathfrak{U}, \mathcal{G}).$$

*Proof.* For each $i$,

$$c_i = \mathbf{e}_{[s_i]_{\mathrm{eq}}}$$

is the basis generator indexed by the strict-equivalence class of $s_i$. For $i < j$ with nonempty overlap,

$$\begin{aligned}(dc)_{ij} &= \rho^{\mathcal{G}}_{U_j U_{ij}}(c_j) - \rho^{\mathcal{G}}_{U_i U_{ij}}(c_i) \\ &= \mathbf{e}_{[\rho_{U_j U_{ij}}(s_j)]_{\mathrm{eq}}} - \mathbf{e}_{[\rho_{U_i U_{ij}}(s_i)]_{\mathrm{eq}}}.\end{aligned}$$

By hypothesis, the two restricted methods are equal. Hence their strict-equivalence classes, and therefore the corresponding basis generators, are equal. The displayed difference consequently vanishes. Thus $dc = 0$, and $c$ is a generator-valued 0-cocycle. □

The proposition shows that exact agreement of local sections is reflected in the abelian coefficient system by the vanishing of the 0-coboundary. The converse is not true: a 0-cocycle, even a generator-valued one, may arise from local sections that are merely strictly equivalent on overlaps rather than literally equal.

The cohomology groups of this complex are abelian invariants attached to the coefficient presheaf $\mathcal{G}$ and the chosen cover $\mathfrak{U}$. The augmented complex

$$\mathcal{G}(U, m_U) \longrightarrow C^0(\mathfrak{U}, \mathcal{G}) \longrightarrow C^1(\mathfrak{U}, \mathcal{G})$$

associates to every compatible 0-cochain $c$ its coset in

$$\frac{\ker\big(C^0(\mathfrak{U}, \mathcal{G}) \to C^1(\mathfrak{U}, \mathcal{G})\big)}{\operatorname{im}\big(\mathcal{G}(U, m_U) \to C^0(\mathfrak{U}, \mathcal{G})\big)}.$$

This coset vanishes exactly when $c$ is the restriction of some element of $\mathcal{G}(U, m_U)$. In particular, if a generator-valued 0-cocycle is obtained by restricting a basis generator

$$\mathbf{e}_{[s]_{\mathrm{eq}}} \in \mathcal{G}(U, m_U)$$

associated with a single global strict-equivalence class, then its coset vanishes. The converse need not hold: an element of $\mathcal{G}(U, m_U)$ mapping to the given family may be a formal integral linear combination rather than a single basis generator.

Thus the augmented quotient gives an abelianized necessary condition for a generator-valued family to arise from the basis generator associated with a single global strict-equivalence class. It does not provide a full obstruction theory for literal strict descent. The coefficient system is generated by strict-equivalence classes rather than retaining the full method structure; a fuller obstruction theory would require a refinement retaining the method data discarded here. Where descent is mediated by nontrivial comparison morphisms, that refinement would generally be nonabelian and adapted to the comparison type used.

# 3. Evaluative Locality and Adjusted Restriction

The evaluative branch is developed in parallel with the strict, lax, and fuzzy descent treatment. This branch concerns methods whose application on a restricted domain may require adjustment before overlap comparison is possible, as in the case of a recipe adapted to a smaller vessel, a proof technique modified to remain valid on a subcase, or a piano fingering revised when a passage is transposed.

In the strict, lax, and fuzzy cases, the central question is how already given local methods compare on overlaps. In the evaluative case, locality may depend on the comparison between direct adjusted restriction and staged adjusted restriction through intermediate domains.

**Adjustment-Data for Restriction**

Let
$$V \hookrightarrow U$$
be an inclusion of domains, and let
$$m_U = (P_{m_U}, S_{m_U}, C_{m_U}, Q_{m_U}, \Theta_{m_U})$$
be a method on $U$. Let
$$A_{UV}(m_U)$$
denote the set of adjustment-data specified for restricting $m_U$ to $V$.

An element
$$a \in A_{UV}(m_U)$$
determines an adjusted restricted method
$$\rho^{\text{adj}}_{UV}(m_U, a) = m_{V,a}$$
on $V$, called the adjusted restriction of $m_U$ to $V$ relative to $a$.

Such adjustment-data may specify modifications of selected step-sequences, configurations, quotient-configurations, structures, or some combination of these, insofar as those modifications are needed for the method to function on the subdomain.

**Induction of Adjustment-Data Under Further Restriction**

Let
$$W \hookrightarrow V \hookrightarrow U$$
be a chain of inclusions, let
$$m_U$$
be a method on $U$, and let
$$a \in A_{UV}(m_U)$$
be adjustment-data determining the adjusted restriction
$$m_{V,a} = \rho^{\text{adj}}_{UV}(m_U, a).$$

An induction operation for adjustment-data under further restriction is a map
$$I_{UVW,m_U} : A_{UV}(m_U) \to A_{UW}(m_U)$$
sending
$$a \longmapsto a|_W := I_{UVW,m_U}(a).$$

The induction operation assigns to adjustment-data for restricting $m_U$ to $V$ a derived adjustment-datum for restricting $m_U$ directly to $W$. It is used to compare direct adjusted restriction from $U$ to $W$ with staged adjusted restriction through the intermediate domain $V$.

The element

$$a|_W \in A_{UW}(m_U)$$

is the derived adjustment-datum on $W$ associated to $a$. Fix provisionally a designated class $\Xi$ of evaluative compatibility types, a formal definition of which is given below.

The induction operation is required to satisfy the following compatibility condition. For every

$$a \in A_{UV}(m_U),$$

there exists at least one

$$b \in A_{VW}(m_{V,a})$$

such that the staged adjusted restriction to $W$,

$$\rho^{\text{adj}}_{VW}(m_{V,a}, b),$$

is evaluatively comparable, relative to $\Xi$, to the direct adjusted restriction determined by the derived adjustment-datum. Thus

$$\rho^{\text{adj}}_{UW}(m_U, a|_W) \approx_\Xi \rho^{\text{adj}}_{VW}(m_{V,a}, b).$$

The choice of $b$ is not required to be unique.

Some methods satisfy a stricter condition, in which exact equality is required for at least one such choice of $b$:

$$\rho^{\text{adj}}_{UW}(m_U, a|_W) = \rho^{\text{adj}}_{VW}(m_{V,a}, b).$$

The induction operation is said to be identity-preserving when, for the identity inclusion

$$U \hookrightarrow U$$

and the identity adjustment-data

$$e_U \in A_{UU}(m_U),$$

one has

$$e_U|_U = e_U.$$

For every chain

$$Z \hookrightarrow W \hookrightarrow V \hookrightarrow U,$$

the induction operation is also required to satisfy the iterated-restriction identity

$$(a|_W)|_Z = a|_Z$$

for every

$$a \in A_{UV}(m_U),$$

whenever both sides are defined by the corresponding induction operations.

### Evaluative Restriction Arrows

The evaluative branch retains the same method-domain objects as the strict branch and equips them with evaluative restriction arrows, adjustment-data, and evaluative compatibility.

An evaluative restriction arrow

$$(V, m_{V,a}) \to (U, m_U)$$

consists of

1. an inclusion of domains
$$V \hookrightarrow U,$$
2. adjustment-data
$$a \in A_{UV}(m_U),$$
3. and the adjusted restricted method
$$m_{V,a} = \rho^{\mathrm{adj}}_{UV}(m_U, a).$$

The additional evaluative burden is carried by the restriction arrow rather than by the object. The method remains the primary object, while the adjustment-data specify how that method is to be taken into the restricted domain. The evaluative branch is therefore developed through evaluative restriction arrows, adjustment-data, and evaluative compatibility, with its main emphasis on the comparison between direct and staged adjusted restriction. Under stronger conditions developed below, this extends to weak composites together with right weak identity.

The evaluative restriction data also include the identity inclusion

$$U \hookrightarrow U$$

together with specified identity adjustment-data

$$e_U \in A_{UU}(m_U),$$

so that same-domain evaluative restriction arrows appear as a special case.

### Evaluative Compatibility Types

Let

$$\Lambda_{\mathrm{eval}}$$

be an index set of evaluative compatibility types, and let

$$\Xi \subseteq \Lambda_{\mathrm{eval}}$$

be a designated class of evaluative compatibility types governing locality in the evaluative branch.

For two adjusted restricted methods on a common domain $W$,

$$m_{W,a} \qquad \text{and} \qquad m_{W,b},$$

and for $\kappa \in \Lambda_{\text{eval}}$, write

$$m_{W,a} \approx_\kappa m_{W,b}$$

when the two adjusted restricted methods are compatible on $W$ in the evaluative sense indexed by $\kappa$.

Write

$$m_{W,a} \approx_\Xi m_{W,b}$$

when there exists some

$$\kappa \in \Xi$$

such that

$$m_{W,a} \approx_\kappa m_{W,b}.$$

The index set $\Lambda_{\text{eval}}$ permits more than one mode of evaluative comparison. The designated class $\Xi$ contains the evaluative compatibility types used for locality in the evaluative branch.

### Raw Evaluative Families

Fix a designated class $\Xi \subseteq \Lambda_{\text{eval}}$ of evaluative compatibility types. Let $(U, m_U)$ be an object.

A raw evaluative family over $(U, m_U)$ is an indexed family

$$\Phi = \{(U_i, m_{U_i,a_i}) \to (U, m_U)\}_{i \in I}$$

such that

$$a_i \in A_{UU_i}(m_U) \qquad \text{and} \qquad m_{U_i,a_i} = \rho^{\text{adj}}_{UU_i}(m_U, a_i)$$

for each $i \in I$.

A raw evaluative family is overlap-defined when, for every pair $i, j \in I$ with nonempty intersection

$$U_{ij} := U_i \cap U_j \neq \varnothing,$$

the overlap $U_{ij}$ is a domain and the induction operation is defined on the adjustment-data $a_i$ and $a_j$, yielding

$$a_i|_{U_{ij}} \in A_{UU_{ij}}(m_U), \qquad a_j|_{U_{ij}} \in A_{UU_{ij}}(m_U)$$

The overlap comparisons are then understood through the corresponding adjusted restrictions determined by these induced adjustment-data on the overlap domain $U_{ij}$.

An overlap-defined raw evaluative family is evaluatively compatible relative to $\Xi$ when, for every pair $i, j \in I$ with nonempty intersection

$$U_{ij} := U_i \cap U_j \neq \varnothing,$$

the adjusted restricted methods

$$m^{(i)}_{U_{ij}} := \rho^{\text{adj}}_{UU_{ij}}(m_U, a_i|_{U_{ij}}) \qquad \text{and} \qquad m^{(j)}_{U_{ij}} := \rho^{\text{adj}}_{UU_{ij}}(m_U, a_j|_{U_{ij}})$$

on $U_{ij}$ satisfy

$$m^{(i)}_{U_{ij}} \approx_\Xi m^{(j)}_{U_{ij}}.$$

An evaluative covering family relative to $\Xi$ of $(U, m_U)$ is an overlap-defined raw evaluative family that is evaluatively compatible relative to $\Xi$ and whose domains jointly cover $U$:

$$U = \bigcup_{i \in I} U_i.$$

The preceding definitions isolate evaluative locality relative to $\Xi$. They provide the local comparison data used in the weak evaluative composition developed next.

#### Weak Evaluative Composability

Let

$$W \hookrightarrow V \hookrightarrow U$$

be inclusions, let

$$a \in A_{UV}(m_U),$$

and let

$$b \in A_{VW}(m_{V,a}), \qquad m_{V,a} = \rho^{\text{adj}}_{UV}(m_U, a).$$

For

$$c \in A_{UW}(m_U),$$

write

$$(a, b) \rightsquigarrow_\Xi c$$

when

$$\rho^{\text{adj}}_{UW}(m_U, c) \approx_\Xi \rho^{\text{adj}}_{VW}(m_{V,a}, b).$$

Thus $c$ is weakly evaluatively composable with the staged adjustment-data $(a, b)$ when the direct adjusted restriction of $m_U$ to $W$ determined by $c$ is evaluatively compatible, relative to $\Xi$, with the staged adjusted restriction through $V$ determined by $a$ and $b$.

**Weak Composites**

Retain the preceding notation. The weak composites of $a$ and $b$ are defined by

$$a *_{\Xi} b := \{ c \in A_{UW}(m_U) : (a,b) \rightsquigarrow_{\Xi} c \}.$$

Thus

$$a *_{\Xi} b$$

is the set of direct adjustment-data on $W$ whose resulting adjusted restrictions are evaluatively compatible, relative to $\Xi$, with the staged adjusted restriction determined by $a$ and $b$. It is not assumed to contain a unique element, and it may be empty.

**Right Weak Identity for Evaluative Composition**

The set-valued weak evaluative composition is said to satisfy right weak identity when, for every inclusion

$$V \hookrightarrow U$$

and every

$$a \in A_{UV}(m_U),$$

with

$$m_{V,a} = \rho^{\mathrm{adj}}_{UV}(m_U, a),$$

and identity adjustment-data

$$e_V \in A_{VV}(m_{V,a}),$$

one has

$$a \in a *_{\Xi} e_V.$$

Thus composition with identity adjustment-data on the restricted domain is required to have the original adjustment-datum among its weak composites on the right.

This is only a right-identity condition. It requires evaluative neutrality of same-domain adjustment on the already adjusted restricted method.

**Proposition — Single-Valued Evaluative Descent Is Available Only for a Singleton Set or Under a Selection Principle**

Let $W \hookrightarrow V \hookrightarrow U$ be inclusions, $a \in A_{UV}(m_U)$, and $b \in A_{VW}(m_{V,a})$. The staged route determines the set

$$a *_{\Xi} b = \{ c \in A_{UW}(m_U) : \rho^{\mathrm{adj}}_{UW}(m_U, c) \approx_{\Xi} \rho^{\mathrm{adj}}_{VW}(m_{V,a}, b) \}$$

of direct adjustment-data, which by definition is not assumed to be a singleton. Where this set contains more than one element, the adjustment-data $a$ and $b$ determine no single direct adjustment-datum $c$ among them. A single-valued descent comparison in the evaluative branch is available only where the set is a singleton by a condition internal to the chosen domain or model, or where a selection principle beyond $a$, $b$, and $\approx_\Xi$ designates one element.

*Proof.* Once $a$ and $b$ are fixed, the staged adjusted restriction $\rho^{\text{adj}}_{VW}(m_{V,a}, b)$ is the determinate adjusted restricted method assigned by $\rho^{\text{adj}}_{VW}$ to $(m_{V,a}, b)$. A direct adjustment-datum $c \in A_{UW}(m_U)$ lies in $a *_\Xi b$, by definition, when its direct adjusted restriction $\rho^{\text{adj}}_{UW}(m_U, c)$ stands to that staged adjusted restriction in the relation $\approx_\Xi$. Admissibility is determined by $\approx_\Xi$ on adjusted restrictions, and depends on $c$ only through whether $\rho^{\text{adj}}_{UW}(m_U, c)$ satisfies that relation with the fixed staged adjusted restriction.

The relation $\approx_\Xi$ holds between two adjusted restricted methods on $W$ when there exists a compatibility type $\kappa \in \Xi$ under which they are compatible. It is a compatibility relation on adjusted restrictions and carries no order, choice function, or distinguished representative. In the present construction, $A_{UW}(m_U)$ is used only as a set of adjustment-data, with no supplied order, choice function, or selection rule. Among the elements of $a *_\Xi b$, each satisfying the same membership condition with respect to the fixed staged adjusted restriction, neither $\approx_\Xi$ nor $A_{UW}(m_U)$ distinguishes one from another.

Where no $c$ lies in $a *_\Xi b$, the staged route has no direct representative. Where exactly one does, that instance is determinate. Where more than one does, the elements stand on equal footing in the sense just established, and a single direct adjustment-datum is obtained only upon a condition internal to the chosen domain or model rendering the set a singleton, or a selection rule beyond the supplied data. $\square$

Once their comparison data are supplied, the strict, lax, and fuzzy branches compare restricted data on overlaps. The evaluative branch yields instead a set of representatives, single-valued only under a further condition or selection.

# 4. Functorial Behavior and Transposition

The evaluative branch shows that restriction may depend on adjustment-data, and that locality may depend on the comparison between staged and direct adjusted restriction. At the level of presheaves of methods, the corresponding question is how restriction-compatible assignments themselves may be compared or transposed.

Fix subsets $K, K' \subseteq \Lambda$. Let $\mathsf{Meth}_K$ be the category whose objects are presheaves of methods

$$\mathcal{M} : \mathsf{MethDom}^{op} \to \mathsf{Struct}_K,$$

and whose morphisms are natural transformations between such presheaves.

A morphism

$$\eta : \mathcal{M} \Rightarrow \mathcal{N}$$

in $\mathsf{Meth}_K$ consists of morphisms

$$\eta_{(U,m_U)} : \mathcal{M}(U, m_U) \to \mathcal{N}(U, m_U)$$

in $\mathsf{Struct}_K$, one for each object $(U, m_U)$, such that for every morphism

$$(V, m_V) \to (U, m_U)$$

one has

$$\eta_{(V,m_V)} \circ \rho^{\mathcal{M}}_{UV} = \rho^{\mathcal{N}}_{UV} \circ \eta_{(U,m_U)}.$$

A structural transposition is a functor

$$\mathsf{T} : \mathsf{Meth}_K \to \mathsf{Meth}_{K'}$$

from one category of method-presheaves to another. It carries a presheaf of methods to a further presheaf of methods, with restriction still defined in the target category.

Structural transposition shifts methods into a new descriptive or operational regime. Some relations, configurations, or procedures of application may be carried forward; others may be weakened, discarded, or replaced. One purpose of the preceding definitions is to determine when methods remain comparable after such a shift.

The naturality condition expresses compatibility with restriction. For each morphism

$$(V, m_V) \to (U, m_U),$$

passing from $\mathcal{M}(U, m_U)$ to $\mathcal{N}(U, m_U)$ and then restricting to $(V, m_V)$ gives the same result as restricting first from $(U, m_U)$ to $(V, m_V)$ and then passing from $\mathcal{M}(V, m_V)$ to $\mathcal{N}(V, m_V)$. A natural transformation between presheaves of methods respects change of domain in this sense.

When structural transposition is expressed by a functor $\mathsf{T} : \mathsf{Meth}_K \to \mathsf{Meth}_{K'}$, the transposed presheaf $\mathsf{T}(\mathcal{M})$ remains organized by domains and restriction maps. The special case $K' = K$ covers transpositions that preserve the same profile of method-morphism kinds before and after the shift.

**Proposition — Structural Transposition with Natural Transformation Preserves Strict Overlap Agreement**

Let

$$\mathsf{T} : \mathsf{Meth}_K \to \mathsf{Meth}_K$$

be a structural transposition. Assume that for each presheaf of methods $\mathcal{M} \in \mathsf{Meth}_K$, there is a natural transformation

$$\tau^{\mathcal{M}} : \mathcal{M} \Rightarrow \mathsf{T}(\mathcal{M}).$$

Let

$$\{(U_i, m_{U_i}) \to (U, m_U)\}$$

be a strict covering family, and let

$$s_i \in \mathcal{M}(U_i, m_{U_i})$$

be local sections satisfying

$$\rho^{\mathcal{M}}_{U_i U_{ij}}(s_i) = \rho^{\mathcal{M}}_{U_j U_{ij}}(s_j) \qquad \text{for every pair } i, j \text{ with nonempty overlap } U_{ij} := U_i \cap U_j.$$

Then the transposed local sections

$$\tau^{\mathcal{M}}_{(U_i, m_{U_i})}(s_i) \in \mathsf{T}(\mathcal{M})(U_i, m_{U_i})$$

also satisfy strict overlap agreement:

$$\rho^{\mathsf{T}(\mathcal{M})}_{U_i U_{ij}}\big(\tau^{\mathcal{M}}_{(U_i, m_{U_i})}(s_i)\big) = \rho^{\mathsf{T}(\mathcal{M})}_{U_j U_{ij}}\big(\tau^{\mathcal{M}}_{(U_j, m_{U_j})}(s_j)\big)$$

for all $i, j$.

*Proof.* Let $U_{ij} = U_i \cap U_j$. Since $\tau^{\mathcal{M}}$ is a natural transformation, one has

$$\rho^{\mathsf{T}(\mathcal{M})}_{U_i U_{ij}} \circ \tau^{\mathcal{M}}_{(U_i, m_{U_i})} = \tau^{\mathcal{M}}_{(U_{ij}, m_{U_{ij}})} \circ \rho^{\mathcal{M}}_{U_i U_{ij}},$$

and similarly

$$\rho^{\mathsf{T}(\mathcal{M})}_{U_j U_{ij}} \circ \tau^{\mathcal{M}}_{(U_j, m_{U_j})} = \tau^{\mathcal{M}}_{(U_{ij}, m_{U_{ij}})} \circ \rho^{\mathcal{M}}_{U_j U_{ij}}.$$

By hypothesis,

$$\rho^{\mathcal{M}}_{U_i U_{ij}}(s_i) = \rho^{\mathcal{M}}_{U_j U_{ij}}(s_j).$$

Applying $\tau^{\mathcal{M}}_{(U_{ij}, m_{U_{ij}})}$ to this equality and using the two naturality identities above gives

$$\rho^{\mathsf{T}(\mathcal{M})}_{U_i U_{ij}}\big(\tau^{\mathcal{M}}_{(U_i, m_{U_i})}(s_i)\big) = \rho^{\mathsf{T}(\mathcal{M})}_{U_j U_{ij}}\big(\tau^{\mathcal{M}}_{(U_j, m_{U_j})}(s_j)\big).$$

The transposed family satisfies strict overlap agreement on the covering family. □

This proposition shows that a structural transposition equipped with a natural transformation $\tau^{\mathcal{M}} : \mathcal{M} \Rightarrow \mathsf{T}(\mathcal{M})$ preserves strict overlap compatibility.

**Combination**

In some domains of inquiry, methods may be combined within a fixed domain in order to maintain, reinforce, or extend a configuration. This can be represented by equipping each $\mathcal{M}(U, m_U)$ with a combination operation

$$\otimes_U : \mathcal{M}(U, m_U) \times \mathcal{M}(U, m_U) \to \mathcal{M}(U, m_U).$$

The value $x \otimes_U y$ is a local method in $U$ obtained by combining the local methods $x$ and $y$. This combination operation is additional to the presheaf data itself. Where such a product is used, restriction is required to be compatible with combination, so that for each morphism

$$(V, m_V) \to (U, m_U),$$

one has

$$\rho_{UV}(x \otimes_U y) = \rho_{UV}(x) \otimes_V \rho_{UV}(y).$$

When every application of the combined local method $x \otimes_U y$ includes an application of $x$, the earlier morphism of extension expresses that inclusion by $x \rightsquigarrow x \otimes_U y$. Similarly, if every application of it includes an application of $y$, this gives $y \rightsquigarrow x \otimes_U y$.

**Proposition — Restriction Stability of Extension under Combination**

Assume that each $\mathcal{M}(U, m_U)$ is equipped with a combination operation

$$\otimes_U : \mathcal{M}(U, m_U) \times \mathcal{M}(U, m_U) \to \mathcal{M}(U, m_U)$$

compatible with restriction, so that for every morphism

$$(V, m_V) \to (U, m_U),$$

one has

$$\rho_{UV}(x \otimes_U y) = \rho_{UV}(x) \otimes_V \rho_{UV}(y).$$

Let $x, y \in \mathcal{M}(U, m_U)$, and let $z$ be $x$ or $y$. Suppose that every application of the combined local method $x \otimes_U y$ includes an application of $z$. Suppose further that this inclusion is stable under restriction in the following sense: for every morphism

$$(V, m_V) \to (U, m_U),$$

whenever the restriction $\rho_{UV}(x \otimes_U y)$ is defined, every application of it includes an application of the restricted local method $\rho_{UV}(z)$. Then

$$z \rightsquigarrow x \otimes_U y$$

and, for every morphism

$$(V, m_V) \to (U, m_U),$$

$$\rho_{UV}(z) \rightsquigarrow \rho_{UV}(x \otimes_U y).$$

*Proof.* By hypothesis, every application of the combined local method $x \otimes_U y$ includes an application of $z$. By the definition of extension, this gives

$$z \rightsquigarrow x \otimes_U y.$$

Now let

$$(V, m_V) \to (U, m_U)$$

be a morphism. By compatibility of restriction with combination,

$$\rho_{UV}(x \otimes_U y) = \rho_{UV}(x) \otimes_V \rho_{UV}(y).$$

By the stated restriction-stability hypothesis, every application of the restricted combined local method $\rho_{UV}(x \otimes_U y)$ includes an application of $\rho_{UV}(z)$. By the definition of extension in the restricted domain $V$, one then has

$$\rho_{UV}(z) \rightsquigarrow \rho_{UV}(x \otimes_U y).$$

The argument draws on the two hypotheses concerning $z$ together with the definition of extension. Each hypothesis concerns the applications of the combined local method $x \otimes_U y$. Every such application includes an application of $z$, on $U$ in the first hypothesis and on $V$ in the second. The definition of extension concerns an inclusion of the same kind. It places the applications of one local method within the applications of another. The combined local method $x \otimes_U y$ stands fixed throughout. The letter $z$ names one of the two local methods entering it. The argument therefore stands for $z$ taken as $x$ and for $z$ taken as $y$. □

This proposition shows that, when each $\mathcal{M}(U, m_U)$ is equipped with a combination operation compatible with restriction, extension relations arising from combined application survive restriction when the included application is preserved.

# 5. Examples and Correspondences

The examples below treat the local-to-global side of the strict, lax, and fuzzy cases.

Example 1 is strict. Its domains are the nonempty subsets of a finite set of territories, and a local section over such a subset is a coloring method on it, with overlap comparison given by equality of restricted sections. The example carries a presheaf of methods on a site of methods, together with the presheaf of those coloring methods whose permitted colors stand free at each territory, which satisfies the strict sheaf condition. It also gives two strictly compatible families for which assembly fails, one in uniqueness and one in existence.

Examples 2A and 2B are lax. In each of them the domain is a planar region $U$ equipped with restricted domains

$$\{U_i \hookrightarrow U\}.$$

A local section over $U_i$ is a method whose selected step-sequences determine a local tessellation on that restricted domain. Overlap comparison is given by structure-preserving quotient-configuration bijections of type sqeq between the quotient-configuration sets of the restricted local methods, preserving the restricted structure maps. Example 2A gives a successful lax descent datum; Example 2B gives a cocycle failure on the triple overlap.

The fuzzy examples concern partial fit in a troubleshooting setting. They show symmetric fuzzy comparison, directional fuzzy overlap comparison, failure of stage-preservation, failure of the triple-overlap law, and a case with common designated structures on the larger domain.

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**Example 1 — Map Coloring**

Let $T$ be a finite set of territories and let $B$ be a set of colors containing at least two members. A domain is a nonempty subset $U \subseteq T$, and an inclusion $V \hookrightarrow U$ is inclusion of subsets.

A step in $U$ is a pair $(x, b)$ with $x \in U$ and $b \in B$: the coloring of the territory $x$ with the color $b$. A configuration in $U$ is a partial coloring of $U$, that is, a map from a subset of $U$ to $B$. The configuration map

$$E_U : \Sigma_U^* \to \mathcal{C}(U)$$

sends a step-sequence to the partial coloring recording, at each territory colored by that sequence, the color placed there last; the empty step-sequence gives the uncolored map, which is the configuration of the empty arrangement in $U$. Write $B^U$ for the complete colorings of $U$. Each complete coloring is designated for continuation, so $B^U \subseteq \mathcal{S}(U)$. Restriction of configurations is restriction of partial colorings:

$$\rho^{\mathcal{C}}_{UV}(c) = c|_V.$$

Let $F \subseteq B^U$ be nonempty. The coloring method $m_F$ on $U$ is given by the selector

$$P_{m_F}(s) = 1 \quad \Longleftrightarrow \quad s \text{ colors each territory of } U \text{ once, and } E_U(s) \in F,$$

with designation $D_{m_F} = F$ and $\delta_{m_F}$ the inclusion. The order in which the territories are colored stands free.

The five components of $m_F$ are then determined by $F$. A selected step-sequence colors each territory once, so the color placed at a territory is the color placed there last, and $E_U(s)$ is a complete coloring lying in $F$; conversely each $f \in F$ is the resulting configuration of the step-sequences coloring the territories of $U$ in some order according to $f$, of which there is at least one since $T$ is finite. Hence

$$C_{m_F} = \{\, E_U(s) : s \in S_{m_F} \,\} = F, \qquad Q_{m_F} = \{\, \mathrm{Quot}_U([s]_{\equiv_U}) : s \in S_{m_F} \,\} = F,$$

the second because $\mathrm{Quot}_U([s]_{\equiv_U}) = E_U(s)$ and $D_{m_F} = C_{m_F}$, and

$$\Theta_{m_F} : F \to \mathcal{S}(U)$$

is the inclusion. The designation has $D_{m_F} = C_{m_F}$, so every designated configuration is produced by the method.

Two coloring methods on $U$ are equal when their classes are equal: the selector, and with it each further component, is recovered from $F$. The assignment $F \mapsto m_F$ is therefore injective, and the classes serve as names for the methods they determine.

The arrangement carried by this example is available in other readings. The territories are places at which one act is taken, the colors are the acts permitted at a place, and $F$ is a class of complete assignments of acts to places. Periods and allotments give a timetable, positions and values give a record, and the results below hold of each reading.

### Restriction of Coloring Methods

For $V \hookrightarrow U$, let $\rho^{\Sigma}_{UV}$ delete from a step-sequence the steps coloring territories outside $V$, and set

$$F|_V = \{\, f|_V : f \in F \,\}, \qquad \rho_{UV}(m_F) = m_{F|_V}.$$

The class $F|_V$ is nonempty, so $m_{F|_V}$ is a coloring method on $V$.

The restriction data satisfy the conditions required of them.

*The selected step-sequences of the restriction are the traces.* Let $s \in S_{m_F}$. The trace $\rho^{\Sigma}_{UV}(s)$ colors each territory of $V$ once, and its resulting configuration is $E_U(s)|_V$, a member of $F|_V$. Hence $\rho^{\Sigma}_{UV}(s) \in S_{m_{F|_V}}$. The selector tests the completed coloring, so deletion of the steps outside $V$ leaves the test standing on what remains.

Every selected step-sequence on $V$ arises in this way. Let $u \in S_{m_{F|_V}}$ with $E_V(u) = g \in F|_V$, and choose $f \in F$ with $f|_V = g$. Adjoining to $u$ the steps coloring the territories of $U \setminus V$ according to $f$, in any order, gives a step-sequence $s$ coloring each territory of $U$ once with $E_U(s) = f \in F$, so $s \in S_{m_F}$, and deletion of the adjoined steps returns $\rho^{\Sigma}_{UV}(s) = u$. Hence $S_{m_{F|_V}}$ is the set of traces of the selected step-sequences of $m_F$, and $m_{F|_V}$ is the method-restriction of $m_F$ in the sense fixed above.

*Compatibility with the configuration maps.* For every step-sequence $s$ in $U$,

$$E_V(\rho^{\Sigma}_{UV}(s)) = E_U(s)|_V = \rho^{\mathcal{C}}_{UV}(E_U(s)),$$

since the color placed last at a territory of $V$ is the same before and after deletion of the steps outside $V$.

*Preservation of configuration-equivalence.* If $E_U(s) = E_U(t)$ then $E_U(s)|_V = E_U(t)|_V$, so $\rho^{\Sigma}_{UV}(s) \equiv_V \rho^{\Sigma}_{UV}(t)$.

*The designation requirement.* Write $m_V$ for the restricted method. For the coloring construction,

$$\{\, E_V(\rho^{\Sigma}_{UV}(s)) : s \in S_{m_F},\ E_U(s) \in D_{m_F} \,\} = \{\, f|_V : f \in F \,\} = F|_V = D_{m_V}.$$

Thus the designation requirement holds, in this case as equality.

*Structure-map compatibility.* Both routes send $f \in Q_{m_F} = F$ to $f|_V$:

$$\Theta_{m_{F|_V}} \circ \rho^{Q}_{UV,m_F} = \rho^{\mathcal{C}}_{UV} \circ \Theta_{m_F}.$$

*Functoriality and identity.* For $W \hookrightarrow V \hookrightarrow U$, deletion of the steps outside $V$ followed by deletion of the steps outside $W$ is deletion of the steps outside $W$, and $(F|_V)|_W = F|_W$; hence $\rho_{VW} \circ \rho_{UV} = \rho_{UW}$ on coloring methods, and likewise for the associated maps on selected step-sequences, quotient-configurations, and structures. Deletion of the steps outside $U$ leaves a step-sequence in $U$ as it stands and $F|_U = F$, so $\rho_{UU}$ is the identity.

### The Site of Territory-Sets

Let $\mathsf{MethDom}_T$ denote the full subcategory of $\mathsf{MethDom}$ whose objects are the pairs $(U, m_{B^U})$ for nonempty $U \subseteq T$, with the arrows given by inclusion of subsets together with the restriction above; the ambient method on $U$ is the coloring method whose class is the whole of $B^U$, and $\rho_{UV}(m_{B^U}) = m_{B^V}$ since $(B^U)|_V = B^V$. The presheaves below are functors on $\mathsf{MethDom}_T$, and the site of this example is that subcategory carrying the covering families now described.

Let $\{U_i\}_{i \in I}$ be nonempty subsets of $U$ with $U = \bigcup_{i \in I} U_i$, and take the family

$$\Phi = \{(U_i, m_{B^{U_i}}) \to (U, m_{B^U})\}_{i \in I}.$$

Each nonempty intersection $U_{ij} = U_i \cap U_j$ is again a nonempty subset of $T$ carrying the corresponding restricted method, so $\Phi$ is overlap-admissible; and

$$\rho_{U_i U_{ij}}(m_{B^{U_i}}) = m_{B^{U_{ij}}} = \rho_{U_j U_{ij}}(m_{B^{U_j}}),$$

so $\Phi$ is strictly compatible. With the covering condition holding by choice, $\Phi$ is a strict covering family. Every nonempty intersection of subsets of $T$ is again a nonempty subset of $T$ carrying its own ambient coloring method, so the nonempty intersections arising in the verification of pairwise overlap, base change, and iterated overlap are again objects of $\mathsf{MethDom}_T$. The domain-closure condition of Section 2 therefore holds here, and with the functoriality laws verified above the covering families meet the hypotheses of the Proposition on Strict Sites.

### The Presheaf of Coloring Methods

Take $K = \{\text{inc}, \text{eq}\}$ and set

$$\mathcal{M}(U, m_{B^U}) = \{\, m_F : \varnothing \neq F \subseteq B^U \,\},$$

with $\mathrm{Hom}^{\text{inc}}(m_F, m_G)$ the sub-method inclusions, present when $F \subseteq G$, and $\mathrm{Hom}^{\text{eq}}(m_F, m_G)$ the strict equivalences, present when $F = G$. Each member is a sub-method of the ambient method, since $S_{m_F} \subseteq S_{m_{B^U}}$.

Restriction carries these classes with it: $F \subseteq G$ gives $F|_V \subseteq G|_V$, and $F = G$ gives $F|_V = G|_V$. Hence $\rho_{UV}$ is a morphism in $\mathsf{Struct}_K$, and with the functoriality and identity laws established above, $\mathcal{M}$ is a presheaf of methods of type $K$.

A class of colorings is territory-wise when the colors available at a territory stand free of the colors placed elsewhere: choosing a color at one territory leaves the choices at the others as they were, and any combination of the available colors is a member of the class. The definition puts that in the following form.

For nonempty $F \subseteq B^U$ and $x \in U$, let

$$F(x) = \{ f(x) : f \in F \} \subseteq B$$

be the color-set of $F$ at $x$, a nonempty subset of $B$. The class $F$ is territory-wise when it is the product of its color-sets:

$$F = \prod_{x \in U} F(x),$$

that is, when $f \in F$ holds for every coloring $f \in B^U$ with $f(x) \in F(x)$ at each territory $x$. Equivalently, $F = \prod_{x \in U} B_x$ for nonempty classes $B_x \subseteq B$, and then $B_x = F(x)$.

The color-sets pass through restriction: for $x \in V$,

$$(F|_V)(x) = \{ f|_V(x) : f \in F \} = F(x).$$

Set

$$\mathcal{M}^{\text{tw}}(U, m_{B^U}) = \{ m_F : \varnothing \neq F \subseteq B^U \text{ and } F \text{ is territory-wise} \},$$

with the same typed classes. Restriction stays within it: the color-sets are nonempty, so

$$\Big( \prod_{x \in U} F(x) \Big) \Big|_V = \prod_{x \in V} F(x),$$

and with the display above this gives $F|_V = \prod_{x \in V} (F|_V)(x)$, so $F|_V$ is territory-wise whenever $F$ is. The ambient method belongs to it, since $B^U = \prod_{x \in U} B$. Hence $\mathcal{M}^{\text{tw}}$ is a presheaf of methods of type $K$ on the same site.

**Proposition — Territory-wise Coloring Methods Satisfy the Strict Sheaf Condition**

The presheaf $\mathcal{M}^{\text{tw}}$ satisfies the strict sheaf condition. That is, let

$$\{(U_i, m_{B^{U_i}}) \to (U, m_{B^U})\}_{i \in I}$$

be a strict covering family, and let

$$s_i = m_{F_i} \in \mathcal{M}^{\text{tw}}(U_i, m_{B^{U_i}})$$

be local sections with

$$\rho_{U_i U_{ij}}(s_i) = \rho_{U_j U_{ij}}(s_j)$$

for every pair $i, j$ with nonempty overlap. Then there is a unique

$$s \in \mathcal{M}^{\text{tw}}(U, m_{B^U})$$

with $\rho_{U U_i}(s) = s_i$ for each $i$.

*Proof.* Each $F_i$ is nonempty and territory-wise, so $F_i = \prod_{x \in U_i} F_i(x)$ with each color-set $F_i(x) \subseteq B$ nonempty.

Let $x \in U_{ij}$. Equality of the restricted sections gives $F_i|_{U_{ij}} = F_j|_{U_{ij}}$, and taking color-sets at $x$ gives

$$F_i(x) = (F_i|_{U_{ij}})(x) = (F_j|_{U_{ij}})(x) = F_j(x).$$

The family $\{U_i\}$ covers $U$, so each $x \in U$ lies in some $U_i$, and the common value

$$B_x := F_i(x) \qquad (x \in U_i)$$

is defined independently of the index chosen. Set

$$F = \prod_{x \in U} B_x,$$

a nonempty territory-wise class with $F(x) = B_x$, and $s = m_F$.

For each $i$,

$$F|_{U_i} = \prod_{x \in U_i} B_x = \prod_{x \in U_i} F_i(x) = F_i,$$

so $\rho_{UU_i}(s) = s_i$.

For uniqueness, let $s' = m_G$ with $G$ nonempty and territory-wise and $\rho_{UU_i}(s') = s_i$ for each $i$. Then $G|_{U_i} = F_i$ for each $i$, and for $x \in U_i$,

$$G(x) = (G|_{U_i})(x) = F_i(x) = B_x.$$

Every $x \in U$ lies in some $U_i$, so $G = \prod_{x \in U} G(x) = \prod_{x \in U} B_x = F$, and $s' = s$. $\square$

The passage from the classes to the methods carries no further condition. The selected step-sequences, the resulting configurations, the quotient-configurations and the structure map of $m_F$ are each determined by $F$, so the assembly of the classes assembles the method-level data with them.

By the Strict Assembly Criterion, a family of territory-wise local sections therefore arises from a global section precisely when the restricted local sections agree on every nonempty overlap, and the global section is then unique. This is the strict sheaf condition holding for a presheaf of methods, on a site of methods, in the sense the model proposes.

**Proposition — Failure of Uniqueness for the Presheaf of Coloring Methods**

Let $T = \{x_1, x_2, x_3\}$, let $B = \{0, 1\}$, and take the strict covering family given by

$$U = T, \qquad U_1 = \{x_1, x_2\}, \qquad U_2 = \{x_2, x_3\}, \qquad U_{12} = \{x_2\}.$$

Let $s_1 = m_{B^{U_1}}$ and $s_2 = m_{B^{U_2}}$ be the ambient local sections. Then there exist distinct

$$t, t' \in \mathcal{M}(U, m_{B^U})$$

with $\rho_{UU_i}(t) = \rho_{UU_i}(t') = s_i$ for $i = 1, 2$. Hence a family of local sections of $\mathcal{M}$ agreeing on every nonempty overlap may have more than one global section.

*Proof.* Both local sections restrict on $U_{12}$ to $m_{B^{U_{12}}}$, so they agree on that overlap.

Take

$$F = B^U, \qquad F' = \{ f \in B^U : f(x_1) \neq f(x_3) \},$$

and set $t = m_F$, $t' = m_{F'}$. These are distinct: any complete coloring $f \in B^U$ with $f(x_1) = f(x_3) = 0$ belongs to $F$ and lies outside $F'$.

For $F$, the restrictions are $B^{U_1}$ and $B^{U_2}$. For $F'$, let $(b_1, b_2) \in B^{U_1}$; with two colors available there is $b_3 \neq b_1$, and the coloring $(b_1, b_2, b_3)$ belongs to $F'$ and restricts to $(b_1, b_2)$. Hence $F'|_{U_1} = B^{U_1}$, and symmetrically $F'|_{U_2} = B^{U_2}$. □

The cover shows $x_1$ and $x_3$ apart, each in the company of $x_2$, and never together. A requirement standing between $x_1$ and $x_3$ leaves the restricted data as they stand, and restriction returns the same local sections whether the requirement is imposed or waived. The class $F'$ stands outside $\mathcal{M}^{\mathrm{tw}}$: its color-sets are $F'(x) = B$ at every territory $x \in U$, so $\prod_{x \in U} F'(x) = B^U$, which holds colorings $F'$ leaves out. The preceding proposition returns uniqueness once the sections are territory-wise.

**Proposition — Failure of Existence for the Presheaf of Coloring Methods**

Let $T = \{x_1, x_2, x_3\}$, let $B = \{0, 1\}$, and take the strict covering family given by

$$U = T, \qquad U_1 = \{x_1, x_2\}, \qquad U_2 = \{x_2, x_3\}, \qquad U_3 = \{x_3, x_1\}.$$

Let

$$F_i = \{ f \in B^{U_i} : f \text{ takes distinct colors at the two territories of } U_i \}, \qquad s_i = m_{F_i}.$$

Then the local sections $s_i$ agree on every overlap, and no $t \in \mathcal{M}(U, m_{B^U})$ satisfies $\rho_{UU_i}(t) = s_i$ for each $i$. Hence the existence part of the strict sheaf condition fails for $\mathcal{M}$.

*Proof.* Each $F_i$ has two members and is nonempty. For $i \neq j$ the overlap $U_i \cap U_j$ is a single territory, and both colors occur there among the two members of $F_i$, so

$$F_i|_{U_i \cap U_j} = B^{U_i \cap U_j} \qquad (i \neq j).$$

The local sections agree on every nonempty overlap.

Suppose $t = m_F$ with $F|_{U_i} = F_i$ for each $i$. Since $F$ is nonempty, choose $f \in F$. Then $f|_{U_i} \in F_i$ for each $i$, so

$$f(x_1) \neq f(x_2), \qquad f(x_2) \neq f(x_3), \qquad f(x_3) \neq f(x_1).$$

With two colors available the first two give $f(x_1) = f(x_3)$, against the third. Hence $F$ is empty. The classes of the members of $\mathcal{M}(U, m_{B^U})$ are nonempty, so the supposition fails, and no $t \in \mathcal{M}(U, m_{B^U})$ restricts to the three local sections. □

The mechanism differs from that of the preceding proposition. There, the local sections leave a global section standing and leave more than one of them standing. Here, each pair of territories carries a requirement met by two colorings, and the three requirements taken together are met by none: the obstruction lies in the passage around the three territories, which no single member of the cover shows. Each $F_i$ stands outside $\mathcal{M}^{\text{tw}}$: its color-sets are $F_i(x) = B$ at each of the two territories $x \in U_i$, so the product $\prod_{x \in U_i} F_i(x)$ has four members while $F_i$ has two.

**Remark — The Classes Range Over All Nonempty Subclasses**

The classes $F$ range over the nonempty subclasses of $B^U$, and the narrower range of classes cut out by requirements between neighboring territories is closed under restriction by no argument. Take $T = \{x_1, x_2, x_3\}$ with $x_2$ neighboring each of $x_1$ and $x_3$, and with $x_1$ and $x_3$ standing apart; let $B = \{0, 1\}$, and let $F$ be the colorings giving neighbors distinct colors. Then $f(x_1) \neq f(x_2)$ and $f(x_2) \neq f(x_3)$ give $f(x_1) = f(x_3)$, so

$$F|_{\{x_1, x_3\}} = \{\, (0,0),\ (1,1) \,\},$$

a class holding two territories to a common color. The trace of a requirement of difference between neighbors is a requirement of agreement between territories the subdomain shows apart. Restriction carries the classes into the full range, and the presheaf is formed over that range.

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**Example 2A — Successful Lax Descent by Consistent Overlap Symmetries**

Let $U$ be a planar domain and let

$$U_1, U_2, U_3 \hookrightarrow U$$

be restricted planar subregions with nonempty triple overlap

$$W = U_1 \cap U_2 \cap U_3.$$

For each $U_i$, let

$$s_i \in \mathcal{M}(U_i, m_{U_i})$$

be a local section determined by a local tessellation method $m_{U_i}$.

A step is the placement of a single regular hexagonal tile along an exposed boundary edge of the partial tessellation, set so that the new tile meets the tile already there with edge-incidence, sharing a whole edge corner to corner. A restricted domain $U_i \hookrightarrow U$ is a subregion of $U$, containing the tiles and the shared edges that lie within it. The selector $P_{m_{U_i}}$ takes value 1 on all and only those finite step-sequences in $\Sigma^*_{U_i}$ that place regular hexagons edge-to-edge in $U_i$, preserve the fixed orientation and side-length of the tessellation, and produce no gap or overlap within the region tessellated so far.

Each $m_{U_i}$ is assumed to be nonempty, so at least one selected tessellation step-sequence exists on $U_i$. Then $S_{m_{U_i}}$ is the set of selected tessellation step-sequences on $U_i$, $C_{m_{U_i}}$ is the set of resulting local tessellation configurations, $Q_{m_{U_i}}$ is the set of quotient-configurations determined by those selected step-sequences, and $\Theta_{m_{U_i}}$ is the corresponding structure map.

In this example, the pairwise overlap comparisons are carried by local symmetries of the tessellation patterns, and the induced quotient-configuration bijections satisfy the cocycle condition on the triple overlap.

Assume that each pairwise overlap properly contains the triple overlap. On the triple overlap $W$, let

$$Q_1^{123} = \{a, b\}, \qquad Q_2^{123} = \{x, y\}, \qquad Q_3^{123} = \{p, q\},$$

with

$$\Theta_1^{123}(a) = T_1, \qquad \Theta_1^{123}(b) = T_2,$$
$$\Theta_2^{123}(x) = T_1, \qquad \Theta_2^{123}(y) = T_2,$$

and

$$\Theta_3^{123}(p) = T_1, \qquad \Theta_3^{123}(q) = T_2.$$

Here $T_1$ and $T_2$ are the local tessellation configurations designated for continuation on the triple overlap. On each pairwise overlap below, the same letters denote configurations on that overlap whose restrictions to $W$ are the two configurations just named; the structure-map compatibility of restriction requires this identification, since each displayed restriction map carries a quotient-configuration assigned to $T_k$ on the pairwise overlap to one assigned to $T_k$ on $W$.

On the pairwise overlap $U_{12} = U_1 \cap U_2$, let

$$Q_1^{12} = \{a_1, a_2, b_1, b_2\}, \qquad Q_2^{12} = \{x_1, x_2, y_1, y_2\},$$

with

$$\Theta_1^{12}(a_1) = T_1, \quad \Theta_1^{12}(a_2) = T_1, \quad \Theta_1^{12}(b_1) = T_2, \quad \Theta_1^{12}(b_2) = T_2,$$
$$\Theta_2^{12}(x_1) = T_1, \quad \Theta_2^{12}(x_2) = T_1, \quad \Theta_2^{12}(y_1) = T_2, \quad \Theta_2^{12}(y_2) = T_2.$$

Let the restriction maps to $W$ be

$$\rho^Q_{U_{12}W, m_1^{12}}(a_1) = a, \quad \rho^Q_{U_{12}W, m_1^{12}}(a_2) = a,$$
$$\rho^Q_{U_{12}W, m_1^{12}}(b_1) = b, \quad \rho^Q_{U_{12}W, m_1^{12}}(b_2) = b,$$
$$\rho^Q_{U_{12}W, m_2^{12}}(x_1) = x, \quad \rho^Q_{U_{12}W, m_2^{12}}(x_2) = x,$$
$$\rho^Q_{U_{12}W, m_2^{12}}(y_1) = y, \quad \rho^Q_{U_{12}W, m_2^{12}}(y_2) = y.$$

Define

$$\alpha_{12}(a_1) = x_1, \quad \alpha_{12}(a_2) = x_2,$$
$$\alpha_{12}(b_1) = y_1, \quad \alpha_{12}(b_2) = y_2.$$

Then $\alpha_{12} : Q_1^{12} \to Q_2^{12}$ is a bijection satisfying

$$\Theta_2^{12} \circ \alpha_{12} = \Theta_1^{12}.$$

On the pairwise overlap $U_{23} = U_2 \cap U_3$, let

$$Q_2^{23} = \{x_1', x_2', y_1', y_2'\}, \qquad Q_3^{23} = \{p_1, p_2, q_1, q_2\},$$

with

$$\begin{aligned}
&\Theta_2^{23}(x_1') = T_1, \quad \Theta_2^{23}(x_2') = T_1, \quad \Theta_2^{23}(y_1') = T_2, \quad \Theta_2^{23}(y_2') = T_2,\\
&\Theta_3^{23}(p_1) = T_1, \quad \Theta_3^{23}(p_2) = T_1, \quad \Theta_3^{23}(q_1) = T_2, \quad \Theta_3^{23}(q_2) = T_2.
\end{aligned}$$

Let the restriction maps to $W$ be

$$\begin{aligned}
&\rho^Q_{U_{23}W,\, m_2^{23}}(x_1') = x, \quad \rho^Q_{U_{23}W,\, m_2^{23}}(x_2') = x,\\
&\rho^Q_{U_{23}W,\, m_2^{23}}(y_1') = y, \quad \rho^Q_{U_{23}W,\, m_2^{23}}(y_2') = y,\\
&\rho^Q_{U_{23}W,\, m_3^{23}}(p_1) = p, \quad \rho^Q_{U_{23}W,\, m_3^{23}}(p_2) = p,\\
&\rho^Q_{U_{23}W,\, m_3^{23}}(q_1) = q, \quad \rho^Q_{U_{23}W,\, m_3^{23}}(q_2) = q.
\end{aligned}$$

Define

$$\begin{aligned}
&\alpha_{23}(x_1') = p_1, \quad \alpha_{23}(x_2') = p_2,\\
&\alpha_{23}(y_1') = q_1, \quad \alpha_{23}(y_2') = q_2.
\end{aligned}$$

Then $\alpha_{23} : Q_2^{23} \to Q_3^{23}$ is a bijection satisfying

$$\Theta_3^{23} \circ \alpha_{23} = \Theta_2^{23}.$$

On the pairwise overlap $U_{13} = U_1 \cap U_3$, let

$$Q_1^{13} = \{a_1', a_2', b_1', b_2'\}, \qquad Q_3^{13} = \{p_1', p_2', q_1', q_2'\},$$

with

$$\begin{aligned}
&\Theta_1^{13}(a_1') = T_1, \quad \Theta_1^{13}(a_2') = T_1, \quad \Theta_1^{13}(b_1') = T_2, \quad \Theta_1^{13}(b_2') = T_2,\\
&\Theta_3^{13}(p_1') = T_1, \quad \Theta_3^{13}(p_2') = T_1, \quad \Theta_3^{13}(q_1') = T_2, \quad \Theta_3^{13}(q_2') = T_2.
\end{aligned}$$

Let the restriction maps to $W$ be

$$\begin{aligned}
&\rho^Q_{U_{13}W,\, m_1^{13}}(a_1') = a, \quad \rho^Q_{U_{13}W,\, m_1^{13}}(a_2') = a,\\
&\rho^Q_{U_{13}W,\, m_1^{13}}(b_1') = b, \quad \rho^Q_{U_{13}W,\, m_1^{13}}(b_2') = b,\\
&\rho^Q_{U_{13}W,\, m_3^{13}}(p_1') = p, \quad \rho^Q_{U_{13}W,\, m_3^{13}}(p_2') = p,\\
&\rho^Q_{U_{13}W,\, m_3^{13}}(q_1') = q, \quad \rho^Q_{U_{13}W,\, m_3^{13}}(q_2') = q.
\end{aligned}$$

Define

$$\alpha_{13}(a_1') = p_1', \quad \alpha_{13}(a_2') = p_2',$$
$$\alpha_{13}(b_1') = q_1', \quad \alpha_{13}(b_2') = q_2'.$$

Then $\alpha_{13} : Q_1^{13} \to Q_3^{13}$ is a bijection satisfying

$$\Theta_3^{13} \circ \alpha_{13} = \Theta_1^{13}.$$

The induced triple-overlap maps are therefore

$$\alpha_{12}^{123}(a) = x, \quad \alpha_{12}^{123}(b) = y,$$
$$\alpha_{23}^{123}(x) = p, \quad \alpha_{23}^{123}(y) = q,$$
$$\alpha_{13}^{123}(a) = p, \quad \alpha_{13}^{123}(b) = q.$$

For $\alpha_{12}$, the restriction square

$$\rho^{Q}_{U_{12}W,\, m_2^{12}} \circ \alpha_{12} = \alpha_{12}^{123} \circ \rho^{Q}_{U_{12}W,\, m_1^{12}}$$

holds because both sides send

$$a_1, a_2 \mapsto x, \qquad b_1, b_2 \mapsto y.$$

The restriction squares for $\alpha_{23}$ and $\alpha_{13}$ are verified analogously.

On the triple overlap, the two routes agree on both elements of $Q_1^{123}$:

$$a \mapsto p, \qquad b \mapsto q.$$

Hence

$$\alpha_{23}^{123} \circ \alpha_{12}^{123} = \alpha_{13}^{123}.$$

This gives a lax descent datum. The pairwise overlap comparisons are structure-preserving quotient-configuration bijections, their restrictions to the triple overlap satisfy the required commutativity squares, and the induced triple-overlap bijections satisfy the cocycle condition.

**Example 2B — Failure of Lax Descent by Cocycle Breakdown**

Let $U$ be a planar domain and let

$$U_1, U_2, U_3 \hookrightarrow U$$

be restricted planar subregions with nonempty triple overlap

$$W = U_1 \cap U_2 \cap U_3.$$

For each $U_i$, let

$$s_i \in \mathcal{M}(U_i, m_{U_i})$$

be a local section determined by a local tessellation method

$$m_{U_i} = (P_{m_{U_i}}, S_{m_{U_i}}, C_{m_{U_i}}, Q_{m_{U_i}}, \Theta_{m_{U_i}}).$$

In this example, a step is the placement of a single tile along an exposed boundary edge of the partial tessellation in such a way that edge-incidence is preserved. The designated relations in $\mathcal{R}(U_i)$ are tile adjacency, edge-incidence, and boundary-matching relations.

This is a lax-descent failure example: overlap comparison is given by structure-preserving quotient-configuration bijections of type sqeq between the quotient-configuration sets of the restricted local methods, preserving the restricted structure maps, but the restricted overlap morphisms fail the cocycle condition on the triple overlap. The comparison is determined by the local tessellation configuration designated for continuation on the overlap patch.

Assume that each pairwise overlap properly contains the triple overlap. On the triple overlap $W$, let the quotient-configuration sets of the restricted local methods be

$$Q_1^{123} = \{a, b\}, \qquad Q_2^{123} = \{x, y\}, \qquad Q_3^{123} = \{p, q\},$$

and let each structure map take the single value $T_1$:

$$\Theta_1^{123}(a) = \Theta_1^{123}(b) = T_1,$$
$$\Theta_2^{123}(x) = \Theta_2^{123}(y) = T_1,$$
$$\Theta_3^{123}(p) = \Theta_3^{123}(q) = T_1.$$

Here $T_1$ is the local tessellation configuration designated for continuation on $W$. The three local methods reach that configuration by distinct local tessellations. Each of them distinguishes two quotient-configurations at it.

This arrangement of the structure maps is what a cocycle failure on the triple overlap requires. Suppose the chosen restricted bijections preserve the structure maps, and suppose the composite $\alpha_{23}^{123} \circ \alpha_{12}^{123}$ differs from the direct bijection $\alpha_{13}^{123}$. Some element of $Q_1^{123}$ then has two distinct images in $Q_3^{123}$. Both images are assigned the structure that element carries, so $\Theta_3^{123}$ takes one value at two distinct quotient-configurations. In Example 2A the corresponding structure map separates $T_1$ from $T_2$, and the cocycle condition holds there of necessity. Here it separates nothing on $W$, and the arrangement of the bijections settles the matter.

On each pairwise overlap below, the letters $T_1$ and $T_2$ denote the configurations designated for continuation on that overlap, and the restriction of each to $W$ is stated where it appears. The restriction maps displayed below are surjective onto the corresponding sets on $W$. By the proposition on a surjective source restriction, each pairwise overlap comparison determines the restricted bijection carried on $W$, so no other selection of restricted bijections is available in this example.

On the pairwise overlap $U_{12} = U_1 \cap U_2$, let

$$Q_1^{12} = \{a_1, a_2, b_1, b_2\}, \qquad Q_2^{12} = \{x_1, x_2, y_1, y_2\},$$

with

$$\Theta_1^{12}(a_1) = T_1, \quad \Theta_1^{12}(a_2) = T_1, \quad \Theta_1^{12}(b_1) = T_2, \quad \Theta_1^{12}(b_2) = T_2,$$
$$\Theta_2^{12}(x_1) = T_1, \quad \Theta_2^{12}(x_2) = T_1, \quad \Theta_2^{12}(y_1) = T_2, \quad \Theta_2^{12}(y_2) = T_2.$$

Here $T_1$ and $T_2$ are the two local tessellation configurations designated for continuation on $U_{12}$. Both restrict to the single configuration $T_1$ on $W$.

Let the restriction maps to $W$ be

$$\begin{aligned}
\rho^Q_{U_{12}W,\, m_1^{12}}(a_1) = a, &\quad \rho^Q_{U_{12}W,\, m_1^{12}}(a_2) = a,\\
\rho^Q_{U_{12}W,\, m_1^{12}}(b_1) = b, &\quad \rho^Q_{U_{12}W,\, m_1^{12}}(b_2) = b,\\
\rho^Q_{U_{12}W,\, m_2^{12}}(x_1) = x, &\quad \rho^Q_{U_{12}W,\, m_2^{12}}(x_2) = x,\\
\rho^Q_{U_{12}W,\, m_2^{12}}(y_1) = y, &\quad \rho^Q_{U_{12}W,\, m_2^{12}}(y_2) = y.
\end{aligned}$$

The map $\rho^Q_{U_{12}W,\, m_1^{12}}$ takes the values $a$ and $b$, which are the two elements of $Q_1^{123}$. The map $\rho^Q_{U_{12}W,\, m_2^{12}}$ takes the values $x$ and $y$, which are the two elements of $Q_2^{123}$. Both maps are surjective onto the corresponding set on $W$.

Define

$$\begin{aligned}
\alpha_{12}(a_1) = x_1, &\quad \alpha_{12}(a_2) = x_2,\\
\alpha_{12}(b_1) = y_1, &\quad \alpha_{12}(b_2) = y_2.
\end{aligned}$$

Then $\alpha_{12} : Q_1^{12} \to Q_2^{12}$ is a bijection satisfying

$$\Theta_2^{12} \circ \alpha_{12} = \Theta_1^{12},$$

since each of $a_1, a_2$ is assigned $T_1$ and sent to an element assigned $T_1$, and each of $b_1, b_2$ is assigned $T_2$ and sent to an element assigned $T_2$.

On the pairwise overlap $U_{23} = U_2 \cap U_3$, let

$$Q_2^{23} = \{x_1', x_2', y_1', y_2'\}, \qquad Q_3^{23} = \{p_1, p_2, q_1, q_2\},$$

and let $U_{23}$ carry a single designated structure $T_1$, so that

$$\Theta_2^{23} \equiv T_1, \qquad \Theta_3^{23} \equiv T_1.$$

Let the restriction maps to $W$ be

$$\begin{aligned}
\rho^Q_{U_{23}W,\, m_2^{23}}(x_1') = x, &\quad \rho^Q_{U_{23}W,\, m_2^{23}}(x_2') = x,\\
\rho^Q_{U_{23}W,\, m_2^{23}}(y_1') = y, &\quad \rho^Q_{U_{23}W,\, m_2^{23}}(y_2') = y,\\
\rho^Q_{U_{23}W,\, m_3^{23}}(p_1) = p, &\quad \rho^Q_{U_{23}W,\, m_3^{23}}(p_2) = p,\\
\rho^Q_{U_{23}W,\, m_3^{23}}(q_1) = q, &\quad \rho^Q_{U_{23}W,\, m_3^{23}}(q_2) = q.
\end{aligned}$$

The map $\rho^{Q}_{U_{23}W,\, m_2^{23}}$ takes the values $x$ and $y$, which are the two elements of $Q_2^{123}$. The map $\rho^{Q}_{U_{23}W,\, m_3^{23}}$ takes the values $p$ and $q$, which are the two elements of $Q_3^{123}$. Both maps are surjective onto the corresponding set on $W$.

Define

$$\alpha_{23}(x_1') = q_1, \quad \alpha_{23}(x_2') = q_2,$$
$$\alpha_{23}(y_1') = p_1, \quad \alpha_{23}(y_2') = p_2.$$

Then $\alpha_{23} : Q_2^{23} \to Q_3^{23}$ is a bijection satisfying

$$\Theta_3^{23} \circ \alpha_{23} = \Theta_2^{23},$$

since both structure maps take the single value $T_1$.

The single designated structure on $U_{23}$ is what permits this crossing. On $U_{13}$ below, the configurations restricting to $p$ and the configurations restricting to $q$ are assigned different structures. A structure map on $U_{23}$ drawing the same distinction would forbid a bijection sending $x$-configurations to $q$-configurations.

On the pairwise overlap $U_{13} = U_1 \cap U_3$, let

$$Q_1^{13} = \{a_1', a_2', b_1', b_2'\}, \qquad Q_3^{13} = \{p_1', p_2', q_1', q_2'\},$$

with

$$\Theta_1^{13}(a_1') = T_1, \quad \Theta_1^{13}(a_2') = T_1, \quad \Theta_1^{13}(b_1') = T_2, \quad \Theta_1^{13}(b_2') = T_2,$$
$$\Theta_3^{13}(p_1') = T_1, \quad \Theta_3^{13}(p_2') = T_1, \quad \Theta_3^{13}(q_1') = T_2, \quad \Theta_3^{13}(q_2') = T_2.$$

As on $U_{12}$, both $T_1$ and $T_2$ restrict to the single configuration $T_1$ on $W$.

Let the restriction maps to $W$ be

$$\rho^{Q}_{U_{13}W,\, m_1^{13}}(a_1') = a, \quad \rho^{Q}_{U_{13}W,\, m_1^{13}}(a_2') = a,$$
$$\rho^{Q}_{U_{13}W,\, m_1^{13}}(b_1') = b, \quad \rho^{Q}_{U_{13}W,\, m_1^{13}}(b_2') = b,$$
$$\rho^{Q}_{U_{13}W,\, m_3^{13}}(p_1') = p, \quad \rho^{Q}_{U_{13}W,\, m_3^{13}}(p_2') = p,$$
$$\rho^{Q}_{U_{13}W,\, m_3^{13}}(q_1') = q, \quad \rho^{Q}_{U_{13}W,\, m_3^{13}}(q_2') = q.$$

The map $\rho^{Q}_{U_{13}W,\, m_1^{13}}$ takes the values $a$ and $b$, which are the two elements of $Q_1^{123}$. The map $\rho^{Q}_{U_{13}W,\, m_3^{13}}$ takes the values $p$ and $q$, which are the two elements of $Q_3^{123}$. Both maps are surjective onto the corresponding set on $W$.

Define

$$\alpha_{13}(a_1') = p_1', \quad \alpha_{13}(a_2') = p_2',$$
$$\alpha_{13}(b_1') = q_1', \quad \alpha_{13}(b_2') = q_2'.$$

Then $\alpha_{13} : Q_1^{13} \to Q_3^{13}$ is a bijection satisfying

$$\Theta_3^{13} \circ \alpha_{13} = \Theta_1^{13}.$$

Each source restriction map displayed above is surjective. By the proposition on a surjective source restriction, each pairwise overlap comparison determines the bijection it restricts to on $W$, and that bijection is obtained by following either side of the restriction square.

For $\alpha_{12}$, the element $a$ has preimage $a_1$, and $\alpha_{12}(a_1) = x_1$ restricts to $x$. The element $b$ has preimage $b_1$, and $\alpha_{12}(b_1) = y_1$ restricts to $y$. Hence

$$\alpha_{12}^W(a) = x, \qquad \alpha_{12}^W(b) = y.$$

For $\alpha_{23}$, the element $x$ has preimage $x_1'$, and $\alpha_{23}(x_1') = q_1$ restricts to $q$. The element $y$ has preimage $y_1'$, and $\alpha_{23}(y_1') = p_1$ restricts to $p$. Hence

$$\alpha_{23}^W(x) = q, \qquad \alpha_{23}^W(y) = p.$$

For $\alpha_{13}$, the element $a$ has preimage $a_1'$, and $\alpha_{13}(a_1') = p_1'$ restricts to $p$. The element $b$ has preimage $b_1'$, and $\alpha_{13}(b_1') = q_1'$ restricts to $q$. Hence

$$\alpha_{13}^W(a) = p, \qquad \alpha_{13}^W(b) = q.$$

Each of these three bijections preserves the structure map on $W$, since every structure map on $W$ takes the single value $T_1$. Taking the second preimage in each case gives the same values, as the proposition requires.

The cocycle condition on $W$ fails. Evaluating at $a$,

$$(\alpha_{23}^W \circ \alpha_{12}^W)(a) = \alpha_{23}^W(x) = q,$$

whereas

$$\alpha_{13}^W(a) = p.$$

Since $p$ and $q$ are distinct,

$$\alpha_{23}^W \circ \alpha_{12}^W \neq \alpha_{13}^W.$$

The three restricted local methods stand compared on the pairwise overlaps by morphisms of type sqeq. Those comparisons determine the bijections carried on the triple overlap, and the determined bijections do not satisfy the cocycle condition. No other selection of restricted bijections is available. The pairwise sqeq-comparisons therefore constitute no lax descent datum, and the corresponding nonabelian Čech 1-cochain fails the cocycle condition. The failure arises before any question of assembly from a lax descent datum, since the datum does not exist on this family.

This example shows that local tessellation methods may stand compared on pairwise overlaps while forming no lax descent datum. Pairwise comparison must also compose consistently on triple overlaps, and here that condition fails.

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**Examples of Fuzzy Comparison and Descent** The fuzzy examples involve a troubleshooting setting. The purpose of the method is to determine the next diagnostic action toward restoring power to a device that will not turn on. In these examples, the values $S_1$ and $S_2$ indicate the next diagnostic stage to which a quotient-configuration leads. On each overlap domain below, the same letters denote designated configurations on that domain; the structure-map compatibility of restriction identifies the stage assigned on the triple overlap with the restriction of the stage assigned on each pairwise overlap.

**A Successful Fuzzy Comparison and Descent Datum** Consider three local methods on pairwise overlaps

$$U_{ij}, \qquad U_{jk}, \qquad U_{ik},$$

with common triple overlap

$$U_{ijk} = U_i \cap U_j \cap U_k,$$

and assume that each pairwise overlap properly contains $U_{ijk}$.

On the triple overlap, let

$$Q_i^{ijk} = \{a, b, c\}, \qquad Q_j^{ijk} = \{x, y\}, \qquad Q_k^{ijk} = \{p, q\},$$

with

$$\Theta_i^{ijk}(a) = S_1, \qquad \Theta_i^{ijk}(b) = S_1, \qquad \Theta_i^{ijk}(c) = S_2,$$
$$\Theta_j^{ijk}(x) = S_1, \qquad \Theta_j^{ijk}(y) = S_2,$$

and

$$\Theta_k^{ijk}(p) = S_1, \qquad \Theta_k^{ijk}(q) = S_2.$$

Interpret $a$ and $b$ as two finer distinctions among external power problems, both leading to the same next diagnostic stage $S_1$, and $c$ as an internal hardware fault leading to the next stage $S_2$.

Now define the quotient-configurations on the pairwise overlaps.

On $U_{ij}$, let

$$Q_i^{ij} = \{a_1, a_2, b, c\}, \qquad Q_j^{ij} = \{x_1, x_2, y\},$$

with

$$\Theta_i^{ij}(a_1) = S_1, \quad \Theta_i^{ij}(a_2) = S_1, \quad \Theta_i^{ij}(b) = S_1, \quad \Theta_i^{ij}(c) = S_2,$$

and

$$\Theta_j^{ij}(x_1) = S_1, \quad \Theta_j^{ij}(x_2) = S_1, \quad \Theta_j^{ij}(y) = S_2.$$

Let restriction to the triple overlap send

$$a_1 \mapsto a, \qquad a_2 \mapsto a, \qquad b \mapsto b, \qquad c \mapsto c$$

on the $i$-side, and

$$x_1 \mapsto x, \qquad x_2 \mapsto x, \qquad y \mapsto y$$

on the $j$-side.

On $U_{jk}$, let

$$Q_j^{jk} = \{x_1', x_2', y'\}, \qquad Q_k^{jk} = \{p_1, p_2, q\},$$

with

$$\Theta_j^{jk}(x_1') = S_1, \quad \Theta_j^{jk}(x_2') = S_1, \quad \Theta_j^{jk}(y') = S_2,$$

and

$$\Theta_k^{jk}(p_1) = S_1, \quad \Theta_k^{jk}(p_2) = S_1, \quad \Theta_k^{jk}(q) = S_2.$$

Let restriction to the triple overlap send

$$x_1' \mapsto x, \qquad x_2' \mapsto x, \qquad y' \mapsto y$$

on the $j$-side, and

$$p_1 \mapsto p, \qquad p_2 \mapsto p, \qquad q \mapsto q$$

on the $k$-side.

On $U_{ik}$, let

$$Q_i^{ik} = \{a_1', a_2', b', c'\}, \qquad Q_k^{ik} = \{p_1', p_2', q'\},$$

with

$$\Theta_i^{ik}(a_1') = S_1, \quad \Theta_i^{ik}(a_2') = S_1, \quad \Theta_i^{ik}(b') = S_1, \quad \Theta_i^{ik}(c') = S_2,$$

and

$$\Theta_k^{ik}(p_1') = S_1, \quad \Theta_k^{ik}(p_2') = S_1, \quad \Theta_k^{ik}(q') = S_2.$$

Let restriction to the triple overlap send

$$a_1' \mapsto a, \qquad a_2' \mapsto a, \qquad b' \mapsto b, \qquad c' \mapsto c$$

on the $i$-side, and

$$p_1' \mapsto p, \qquad p_2' \mapsto p, \qquad q' \mapsto q$$

on the $k$-side.

Now define pairwise fuzzy overlap comparisons by

$$\beta_{ij}^Q(a_1) = x_1, \qquad \beta_{ij}^Q(a_2) = x_1, \qquad \beta_{ij}^Q(b) = x_2, \qquad \beta_{ij}^Q(c) = y,$$

$$\beta_{jk}^Q(x_1') = p_1, \qquad \beta_{jk}^Q(x_2') = p_2, \qquad \beta_{jk}^Q(y') = q,$$

and

$$\beta_{ik}^Q(a_1') = p_1', \qquad \beta_{ik}^Q(a_2') = p_1', \qquad \beta_{ik}^Q(b') = p_2', \qquad \beta_{ik}^Q(c') = q'.$$

These preserve the configurations designated for continuation:

$$\Theta_j^{ij} \circ \beta_{ij}^Q = \Theta_i^{ij}, \qquad \Theta_k^{jk} \circ \beta_{jk}^Q = \Theta_j^{jk}, \qquad \Theta_k^{ik} \circ \beta_{ik}^Q = \Theta_i^{ik}.$$

Each pairwise fuzzy overlap comparison also yields a symmetric fuzzy comparison on its overlap: stage preservation gives $\Theta_i^{ij}(Q_i^{ij}) \subseteq \Theta_j^{ij}(Q_j^{ij})$, and these classes are nonempty, so the two restricted local methods share at least one designated structure there.

Define the corresponding triple-overlap maps by

$$\beta_{ij}^{Q,ijk}(a) = x, \qquad \beta_{ij}^{Q,ijk}(b) = x, \qquad \beta_{ij}^{Q,ijk}(c) = y,$$

$$\beta_{jk}^{Q,ijk}(x) = p, \qquad \beta_{jk}^{Q,ijk}(y) = q,$$

and

$$\beta_{ik}^{Q,ijk}(a) = p, \qquad \beta_{ik}^{Q,ijk}(b) = p, \qquad \beta_{ik}^{Q,ijk}(c) = q.$$

Now verify the restriction-square condition.

For $\beta_{ij}^{Q}$, the two sides of the restriction square send

$$a_1 \mapsto x, \qquad a_2 \mapsto x, \qquad b \mapsto x, \qquad c \mapsto y.$$

For $\beta_{jk}^{Q}$, the two sides of the restriction square send

$$x_1' \mapsto p, \qquad x_2' \mapsto p, \qquad y' \mapsto q.$$

For $\beta_{ik}^{Q}$, the two sides of the restriction square send

$$a_1' \mapsto p, \qquad a_2' \mapsto p, \qquad b' \mapsto p, \qquad c' \mapsto q.$$

Thus each pairwise fuzzy overlap comparison restricts to the corresponding triple-overlap map in the sense of the definition.

On the triple overlap, the two routes agree on all elements of $Q_i^{ijk}$:

$$a \mapsto p, \qquad b \mapsto p, \qquad c \mapsto q.$$

Hence

$$\beta_{jk}^{Q,ijk} \circ \beta_{ij}^{Q,ijk} = \beta_{ik}^{Q,ijk}.$$

This gives a fuzzy descent datum with nontrivial restriction from the pairwise overlaps to the triple overlap. The finer local distinctions available on the larger pairwise overlaps are partially identified under restriction to the smaller common overlap, and the triple-overlap law is satisfied after that restriction.

**Stage-Preservation for Fuzzy Overlap Maps** The stage-preservation condition for a fuzzy overlap map excludes apparent counterexamples such as the following: with the first and third local methods as above, the attempted map

$$\beta_{ik}^{Q,ijk}(a) = p, \qquad \beta_{ik}^{Q,ijk}(b) = q, \qquad \beta_{ik}^{Q,ijk}(c) = q$$

fails the stage-preservation condition, as the following shows.

The defining condition for a fuzzy overlap comparison is the stage-preservation condition

$$\Theta_k^{ijk} \circ \beta_{ik}^{Q,ijk} = \Theta_i^{ijk}.$$

At $a$,

$$(\Theta_k^{ijk} \circ \beta_{ik}^{Q,ijk})(a) = \Theta_k^{ijk}(p) = S_1 = \Theta_i^{ijk}(a).$$

At $b$,

$$(\Theta_k^{ijk} \circ \beta_{ik}^{Q,ijk})(b) = \Theta_k^{ijk}(q) = S_2,$$

whereas

$$\Theta_i^{ijk}(b) = S_1.$$

The stage-preservation condition breaks at $b$. A fuzzy overlap map satisfying this condition preserves the configuration designated for continuation pointwise, so quotient-configurations assigned to $S_1$ remain assigned to $S_1$, and quotient-configurations assigned to $S_2$ remain assigned to $S_2$. A failure of the triple-overlap law can only be tested after the maps under comparison satisfy stage preservation.

**Failure of the Triple-Overlap Law** Define the third local method so that it distinguishes two different quotient-configurations to which the same stage $S_1$ is assigned. Let

$$Q_k^{ijk} = \{p, r, q\},$$

with

$$\Theta_k^{ijk}(p) = S_1, \qquad \Theta_k^{ijk}(r) = S_1, \qquad \Theta_k^{ijk}(q) = S_2.$$

Let

$$Q_i^{ijk} = \{a, b, c\}, \qquad \Theta_i^{ijk}(a) = S_1, \quad \Theta_i^{ijk}(b) = S_1, \quad \Theta_i^{ijk}(c) = S_2,$$

and let

$$Q_j^{ijk} = \{x, y, z\},$$

with

$$\Theta_j^{ijk}(x) = S_1, \qquad \Theta_j^{ijk}(y) = S_1, \qquad \Theta_j^{ijk}(z) = S_2.$$

On the pairwise overlap $U_{ij}$, let

$$Q_i^{ij} = \{a_1, a_2, b_1, c_1\}, \qquad Q_j^{ij} = \{x_1, x_2, y_1, z_1\},$$

with $a_1, a_2, b_1, x_1, x_2, y_1$ assigned $S_1$ and $c_1, z_1$ assigned $S_2$, and let restriction to $U_{ijk}$ send

$$a_1 \mapsto a, \quad a_2 \mapsto a, \quad b_1 \mapsto b, \quad c_1 \mapsto c,$$

$$x_1 \mapsto x, \quad x_2 \mapsto x, \quad y_1 \mapsto y, \quad z_1 \mapsto z.$$

Define

$$\beta^Q_{ij}(a_1) = x_1, \qquad \beta^Q_{ij}(a_2) = x_2, \qquad \beta^Q_{ij}(b_1) = y_1, \qquad \beta^Q_{ij}(c_1) = z_1.$$

On the pairwise overlap $U_{jk}$, let

$$Q^{jk}_j = \{x'_1, y'_1, z'_1\}, \qquad Q^{jk}_k = \{p'_1, r'_1, q'_1\},$$

with $x'_1, y'_1, p'_1, r'_1$ assigned $S_1$ and $z'_1, q'_1$ assigned $S_2$, and let restriction to $U_{ijk}$ send

$$x'_1 \mapsto x, \quad y'_1 \mapsto y, \quad z'_1 \mapsto z, \qquad p'_1 \mapsto p, \quad r'_1 \mapsto r, \quad q'_1 \mapsto q.$$

Define

$$\beta^Q_{jk}(x'_1) = p'_1, \qquad \beta^Q_{jk}(y'_1) = p'_1, \qquad \beta^Q_{jk}(z'_1) = q'_1.$$

On the pairwise overlap $U_{ik}$, let

$$Q^{ik}_i = \{a''_1, b''_1, c''_1\}, \qquad Q^{ik}_k = \{p''_1, r''_1, q''_1\},$$

with $a''_1, b''_1, p''_1, r''_1$ assigned $S_1$ and $c''_1, q''_1$ assigned $S_2$, and let restriction to $U_{ijk}$ send

$$a''_1 \mapsto a, \quad b''_1 \mapsto b, \quad c''_1 \mapsto c, \qquad p''_1 \mapsto p, \quad r''_1 \mapsto r, \quad q''_1 \mapsto q.$$

Define

$$\beta^Q_{ik}(a''_1) = p''_1, \qquad \beta^Q_{ik}(b''_1) = r''_1, \qquad \beta^Q_{ik}(c''_1) = q''_1.$$

Each of these three comparisons preserves the assigned stage, since every configuration assigned $S_1$ is sent to a configuration assigned $S_1$, and likewise for $S_2$. On $U_{ij}$ the restriction of the source set takes the values $a$, $b$, and $c$, which are the three elements of $Q^{ijk}_i$. On $U_{jk}$ it takes the values $x$, $y$, and $z$, which are the three elements of $Q^{ijk}_j$. On $U_{ik}$ it takes the values $a$, $b$, and $c$, which are again the three elements of $Q^{ijk}_i$. Each of these three restrictions is surjective. By the proposition on a surjective source restriction, each pairwise comparison determines the map it restricts to on $U_{ijk}$, and following either side of the restriction square gives the maps displayed next.

Define

$$\beta^{Q,ijk}_{ij}(a) = x, \qquad \beta^{Q,ijk}_{ij}(b) = y, \qquad \beta^{Q,ijk}_{ij}(c) = z.$$

This preserves the assigned stage, since

$$\Theta^{ijk}_j \circ \beta^{Q,ijk}_{ij} = \Theta^{ijk}_i.$$

Define

$$\beta_{jk}^{Q,ijk}(x) = p, \qquad \beta_{jk}^{Q,ijk}(y) = p, \qquad \beta_{jk}^{Q,ijk}(z) = q.$$

Again,

$$\Theta_k^{ijk} \circ \beta_{jk}^{Q,ijk} = \Theta_j^{ijk}.$$

Now define the direct map

$$\beta_{ik}^{Q,ijk}(a) = p, \qquad \beta_{ik}^{Q,ijk}(b) = r, \qquad \beta_{ik}^{Q,ijk}(c) = q.$$

This too preserves the assigned stage, because both $p$ and $r$ are assigned to $S_1$:

$$\Theta_k^{ijk} \circ \beta_{ik}^{Q,ijk} = \Theta_i^{ijk}.$$

All three pairwise fuzzy maps satisfy stage-preservation.

The two routes agree at $a$ and $c$, but differ at $b$:

$$a \mapsto p, \qquad c \mapsto q,$$

while

$$(\beta_{jk}^{Q,ijk} \circ \beta_{ij}^{Q,ijk})(b) = p, \qquad \beta_{ik}^{Q,ijk}(b) = r.$$

Thus

$$\beta_{jk}^{Q,ijk} \circ \beta_{ij}^{Q,ijk} \neq \beta_{ik}^{Q,ijk}.$$

This is a failure of the triple-overlap law. All three maps preserve the assigned stages. The direct route from the first local method to the third distinguishes between two quotient-configurations assigned to $S_1$, while the indirect route through the second local method identifies them. The triple-overlap law imposes path-independence on stage-preserving transport of quotient-configurations between overlaps.

The three triple-overlap maps are determined by the pairwise fuzzy overlap comparisons exhibited above, since each source restriction is surjective. No other selection of triple-overlap maps is available for those comparisons. The family of pairwise fuzzy overlap comparisons therefore constitutes no fuzzy descent datum.

**A Case with Common Designated Structures on the Larger Domain** The preceding examples show that pairwise fuzzy comparison does not by itself settle the triple-overlap law. In the next example, the designated structures of the local methods arise by restriction from a single pair of structures on the larger domain $U$.

Fix two structures

$$S_1, S_2 \in \mathcal{S}(U),$$

the configurations designated for continuation toward the external-power and internal-hardware diagnoses respectively. On each restricted domain below, the same letters denote the restrictions of these two structures to that domain.

Now let three local methods articulate these common stages in different ways.

Let $m_{U_i}$ be a local method on $U_i$ with

$$Q_{m_{U_i}} = \{a, b, c\},$$

and

$$\Theta_{m_{U_i}}(a) = S_1, \qquad \Theta_{m_{U_i}}(b) = S_1, \qquad \Theta_{m_{U_i}}(c) = S_2.$$

Let $m_{U_j}$ be a local method on $U_j$ with

$$Q_{m_{U_j}} = \{x, y\},$$

and

$$\Theta_{m_{U_j}}(x) = S_1, \qquad \Theta_{m_{U_j}}(y) = S_2.$$

Let $m_{U_k}$ be a local method on $U_k$ with

$$Q_{m_{U_k}} = \{p, r, q\},$$

and

$$\Theta_{m_{U_k}}(p) = S_1, \qquad \Theta_{m_{U_k}}(r) = S_1, \qquad \Theta_{m_{U_k}}(q) = S_2.$$

Each structure map takes its values among the two fixed stages:

$$\Theta_{m_{U_i}}(Q_{m_{U_i}}) = \{S_1, S_2\}, \qquad \Theta_{m_{U_j}}(Q_{m_{U_j}}) = \{S_1, S_2\}, \qquad \Theta_{m_{U_k}}(Q_{m_{U_k}}) = \{S_1, S_2\},$$

each equality read on the domain of the method concerned, the letters denoting the restrictions fixed above. The three local methods therefore designate, on their respective domains, the restrictions of a single pair of structures on $U$.

The common pair of stages is articulated differently by the three local methods: $m_{U_i}$ and $m_{U_k}$ each distinguish two quotient-configurations assigned to $S_1$ and one assigned to $S_2$, while $m_{U_j}$ distinguishes exactly one quotient-configuration for each stage.

**Proposition — Structure-Preserving Quotient Comparison Does Not Imply Strict Equality**

In general, a method-comparison of type sqeq does not imply a method-comparison of type eq.

*Proof.* Let $U$ be a domain in which two distinct step-sequences $s \neq t$ determine the same resulting configuration,

$$E_U(s) = E_U(t) = c,$$

for instance two orders of placement of the same tiles determining the same local tessellation configuration. Let $(D, \delta)$ be a designation with $c \in D$. Define methods $m_U$ and $n_U$ by

$$S_{m_U} = \{s\}, \qquad S_{n_U} = \{t\},$$

each taken with the designation $(D, \delta)$. Then $S_{m_U} \neq S_{n_U}$, so $m_U$ and $n_U$ are not related by a comparison of type eq. Since $s \equiv_U t$, both methods determine the single quotient-configuration

$$Q_{m_U} = \{\mathrm{Quot}_U([s]_{\equiv_U})\} = Q_{n_U},$$

and the identity bijection $\phi_Q$ satisfies $\Theta_{n_U} \circ \phi_Q = \Theta_{m_U}$, both structure maps assigning $\delta(c)$. Thus $m_U \cong_{SQ} n_U$ while $m_U \not\cong n_U$. □

**Summary**

In the examples above, methods are applied locally within restricted domains of a larger domain, and their results are compared on overlaps. In the strict case, overlap comparison is equality of restricted sections. In the lax case, overlap comparison is given by structure-preserving quotient-configuration bijections between the quotient-configuration sets of restricted local methods, preserving the restricted structure maps. In the fuzzy case, comparison is first given symmetrically by shared designated structures on an overlap and, in the directional case, by fuzzy overlap comparisons satisfying a triple-overlap law. The directional case maps from one local quotient-configuration set to another, each underlying a comparison of the restricted local methods themselves. On each triple overlap the datum carries chosen maps restricting them, and those chosen maps are composed and tested by the triple-overlap law. These comparison conditions test the coherence of local data on overlaps.

Assembly is a further matter. Example 1 exhibits a presheaf of methods on which the strict sheaf condition holds, so that a family of local sections there agreeing on every nonempty overlap has one global section and no other. The two propositions following it show that the condition may fail on a presheaf of methods: one gives a family of local sections agreeing on every nonempty overlap with more than one global section, and the other gives such a family with none. Under partial fit, the comparison sends each quotient-configuration of the first restricted method to one of the second that determines the same structure. That map is a morphism of type sqmap from the first restricted method to the second. The comparison preserves the structures, and the map that carries them compares the two restricted methods.

Where strict equality or weaker overlap compatibility fails, the mismatch identifies incompatible local regimes. In Example 2B, those regimes appear as pairwise overlap symmetries that do not satisfy the cocycle condition on the triple overlap. Refining domains, distinctions, or representations may alter the overlap data and change the possibility of global assembly.

# Conclusion

This paper develops a sheaf-theoretic analysis of methods through locality, restriction, comparison on overlaps, and descent. In the strict case, the functorial restriction and domain-closure hypotheses yield a coverage determining a Grothendieck topology, and exact agreement of restricted local methods gives the strict comparison condition.

Section 5 exhibits a presheaf of coloring methods on which the strict sheaf condition holds, and two families of coloring methods agreeing on every nonempty overlap, one having more than one global section and the other having none. In the lax and fuzzy cases, exact agreement is replaced by weaker overlap comparison: in the lax case by structure-preserving quotient-configuration bijections, and in the fuzzy case by shared designated structures on overlaps together with directional fuzzy overlap maps and their triple-overlap law. In both cases, local comparison is tested on overlaps and on triple overlaps.

The lax comparisons support a nonabelian Čech viewpoint, and the fuzzy branch supplies directional overlap maps whose restrictions satisfy a triple-overlap law equating composite and direct restricted comparisons. In the strict case, an abelianization of the coefficients gives a Čech complex of abelian groups, whose augmented quotient furnishes a necessary condition for a family of local sections to arise from a single global class.

The strict, lax, and fuzzy branches compare already given local methods on overlaps. The evaluative branch isolates a different difficulty: restriction itself may require adjustment-data, and locality may depend on comparison between direct adjusted restriction and staged adjusted restriction through intermediate domains. Accordingly, the evaluative case is developed in parallel through weak composites and right weak identity. It extends the earlier descent picture to cases in which restriction requires adjustment before comparison.

The functorial results show what remains available under structural transposition and under combination compatible with restriction. Structural transposition preserves strict local compatibility under a natural transformation, and combination shows how extension relations may survive restriction when combined application is stable.

The examples separate the cases. A strictly compatible family may determine a single global section, or more than one, or none; lax overlap symmetries may succeed or fail according to the cocycle condition; and a fuzzy comparison carries the quotient-configurations of one restricted method to those of another that determine the same structures, comparing the two methods under partial fit.

Taken together, the strict, lax, fuzzy, and evaluative cases distinguish strict compatibility, weakened overlap comparison, and set-valued evaluative descent. The abelian Čech complex of the strict case furnishes a necessary condition for a family of local sections to arise from a single global class, and the nonabelian lax and directional fuzzy viewpoints indicate the form a fuller obstruction theory of methods would take.